

ME5412 Robotics for Healthcare

Interaction Control

交互控制

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Outline

- Recalling robot control basics
 - Introducing external forces in the robot dynamic model
 - Techniques for controlling interaction forces
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- 回顾机器人控制基础
 - 在机器人动力学模型中引入外力
 - 交互力控制技术

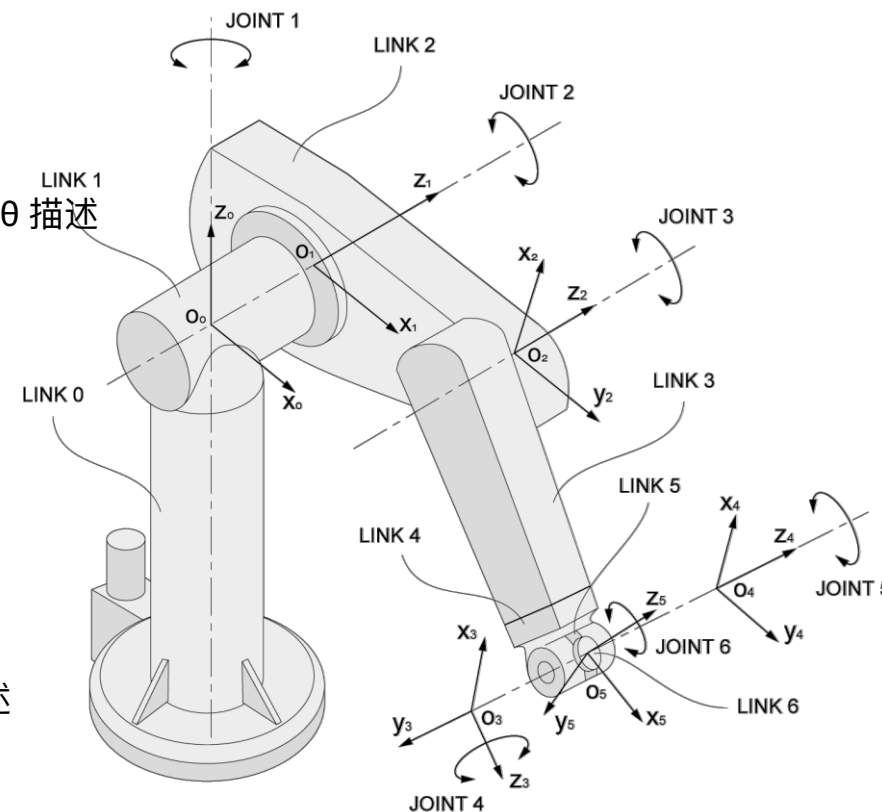
经典机械臂

A Classical Robotic Arm

- 刚性连杆串联而成
- 基座固定，自由端为末端执行器
- 连杆通过旋转关节连接
 - 拟人臂结构
- A chain of rigid links
- The base is fixed, the free end is the end effector
- Links connected with revolute joints
 - Anthropomorphic arm
 - 旋转关节（转动关节）
- Rotational joint (revolute)
 - 两个连杆的相对位置完全由一个角度 θ 描述
 - The relative position of 2 links is fully described by an angle θ

Other joint type:

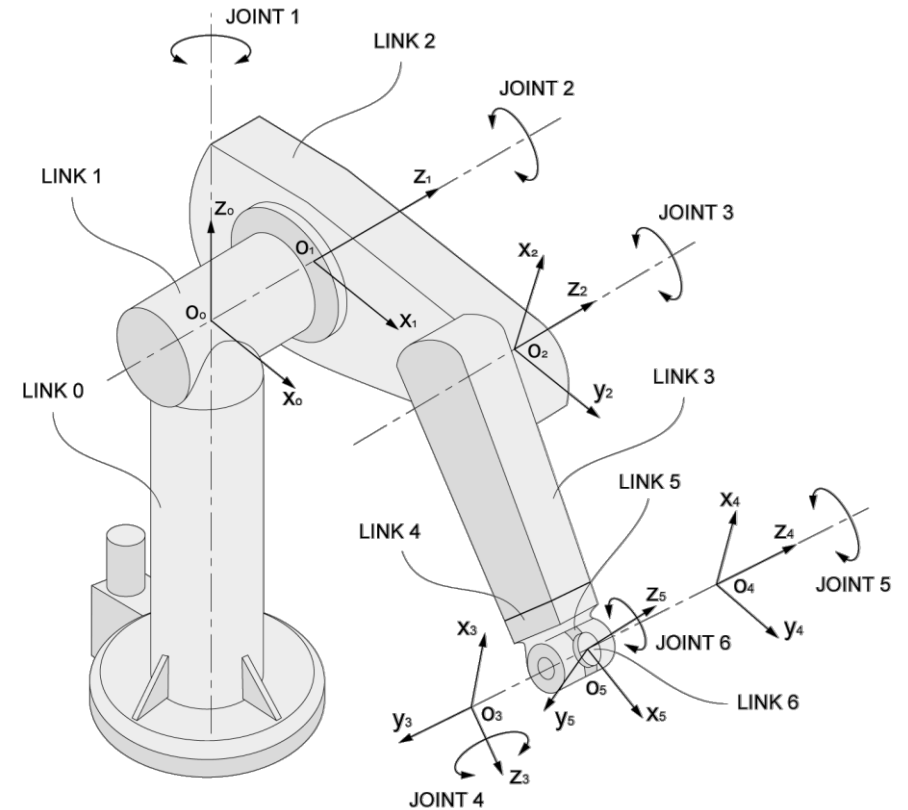
- Translational joint (prismatic)
 - The relative position of 2 links is fully described by a distance d
 - If all prismatic joints
 - Cartesian robot
- 另一种关节类型：
- 移动关节（平移关节）
 - 两个连杆的相对位置完全由一个距离 d 描述
 - 如果全部是移动关节
 - 直角坐标机器人



自由度

Degree of Freedom (DOF)

- 自由度 (DOF) 是完全描述一个机械系统运动所需的最少坐标数中的任意一个。
- 如图所示的机器人，连杆-关节对的数量决定了自由度的数量。
- 假设每个关节都配有一个执行器（如电机），且关节的旋转轴与电机轴线重合。
- The **degree of freedom (DOF)** is any of the minimum number of coordinates required to completely specify the motion of a mechanical system.
- In a robot like the one in the figure, the number of link-joint pairs determines the number of DOFs
- We assume we have an **actuator** (e.g., motor) on each joint and that the rotation axes of the joint and the motor coincide.



关节空间与笛卡尔空间

Joint Space and Cartesian Space

- 关节空间（或称构型空间）：定义关节变量向量 $\mathbf{q}$ 所在的空间。

- 关节空间维数记为 n (n = 机器人关节数)。

- **Joint space** (or configuration space) is the space in which the $\mathbf{q}$ vector of joint variables is defined.

- The joint space dimension is indicated with n (n = number of joints in the robot).

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \vdots \\ q_{n-1} \end{bmatrix}$$

- **Cartesian space** (or task space) is the space in which the

$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \Phi \end{bmatrix}$ vector of the end-effector position is defined.

- The Cartesian space dimension is indicated with m ($m = 6$).

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \Phi \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ \phi \\ \theta \\ \psi \end{bmatrix}$$

- 笛卡尔空间（或称任务空间）：定义末端执行器位置向量 $\mathbf{x}$ 所在的空间。

- 笛卡尔空间维数记为 m ($m = 6$)。

正（直接）运动学

Forward (or Direct) Kinematics

- Given a position $\mathbf{q}$ in joint space, the corresponding position in Cartesian space $\mathbf{x}$ is uniquely determined
 - 给定关节空间中的位置 $\mathbf{q}$ ，笛卡尔空间中对应的位置 $\mathbf{x}$ 被唯一确定。
- The function to obtain $\mathbf{x}$ from $\mathbf{q}$ is the forward kinematic transformation K :
 - 由 $\mathbf{q}$ 求 $\mathbf{x}$ 的函数即为正运动学变换 K ：

$$\mathbf{x} = K(\mathbf{q})$$

$\mathbf{x} = K(\mathbf{q})$

Inverse Kinematics 逆运动学

- Given a desired position $\mathbf{x}$ in Cartesian space, the corresponding position in joint space is given by the inverse kinematic transformation K^{-1} :
 - 给定笛卡尔空间中的期望位置 $\mathbf{x}$ ，关节空间中对应的位置由逆运动学变换 K^{-1} 给出：

$$\mathbf{q} = K^{-1}(\mathbf{x})$$

$\mathbf{q} = K^{-1}(\mathbf{x})$

微分运动学

Differential Kinematics

- 速度空间中的变换。
- 正向微分运动学：从关节速度到末端执行器速度：

- Transformations in the velocity space.
- **Forward differential kinematics**, from joint velocity to end-effector velocity:

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

- **Inverse differential kinematics**, from end-effector velocity to joint velocity:

- 逆向微分运动学：从末端执行器速度到关节速度：

$$\dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\dot{\mathbf{x}}$$

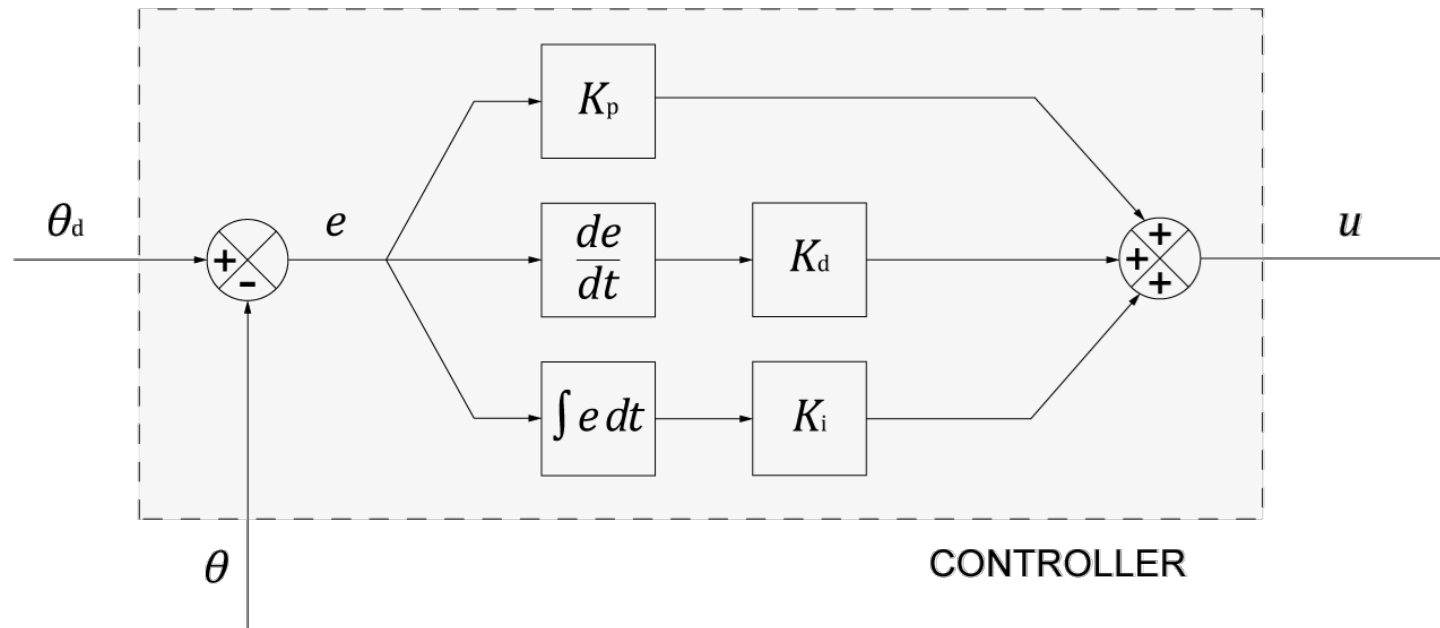
- 变换矩阵即为雅可比矩阵 $\mathbf{J}_{\mathbf{q}}$ 。
- The transformation matrix is the Jacobian matrix $\mathbf{J}(\mathbf{q})$
- The Jacobian matrix is not composed of geometric, constant parameters but depends on the current robot configuration $\mathbf{q}$.

雅可比矩阵并非由几何常量参数组成，而是取决于当前机器人构型 $\mathbf{q}$ 。

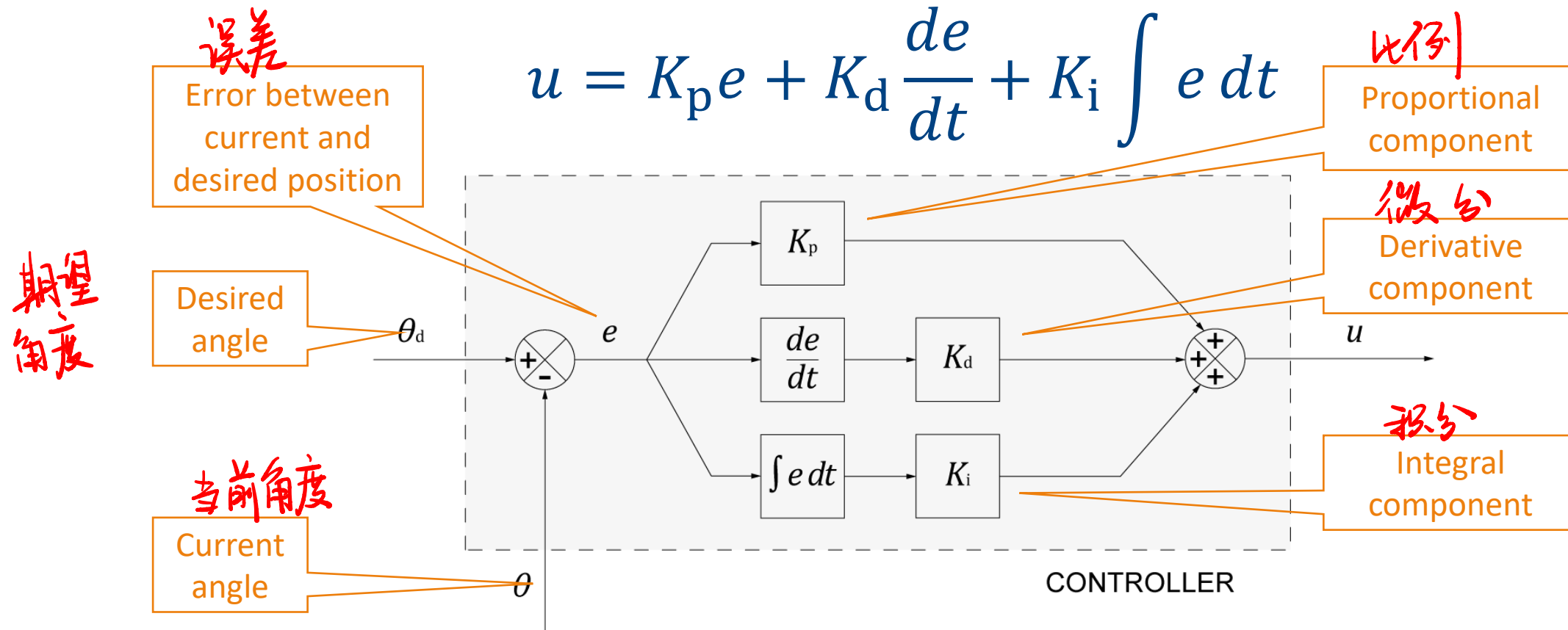
PID (Proportional, Integral, Derivative) Control

PID (比例-积分-微分) 控制

$$u = K_p e + K_d \frac{de}{dt} + K_i \int e dt$$



PID (Proportional, Integral, Derivative) Control

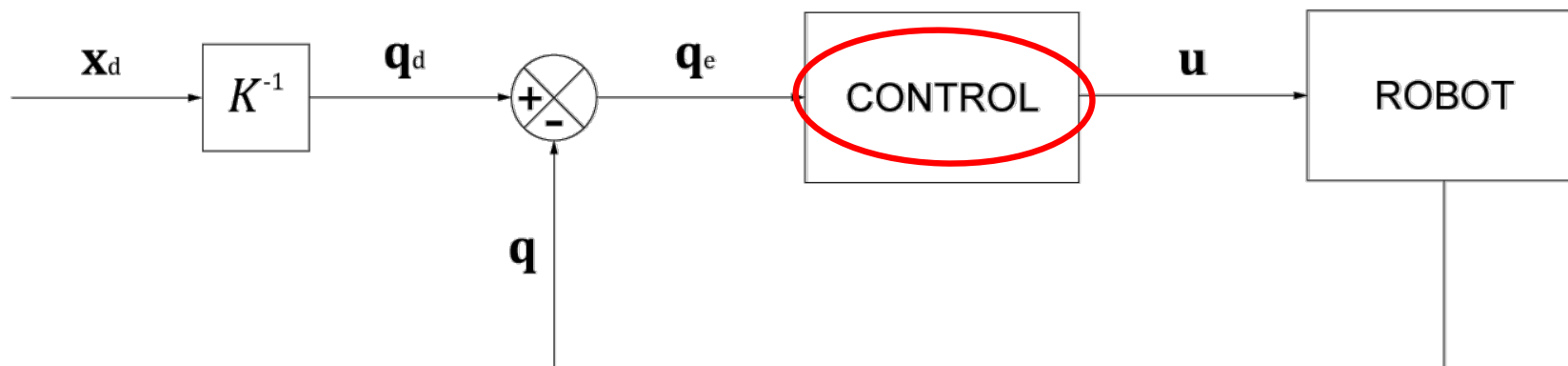


关节空间控制*

Joint Space Control*

- Inverse kinematic transformation K^{-1} at the start of the control system, outside of the control loop.
- desired position $\mathbf{x}_d$ in task space $\rightarrow$ desired position $\mathbf{q}_d$ in joint space.
- $\mathbf{q}_d$ is given by position sensors.
- The error $\mathbf{q}_e$ is in joint space.

- 逆运动学变换 K^{-1} 在控制系统入口处执行，位于控制回路之外。
- 任务空间中的期望位置 $\mathbf{x}_d \rightarrow$ 关节空间中的期望位置 $\mathbf{q}_d$ 。
- $\mathbf{q}_d$ 由位置传感器给出。
- 误差 $\mathbf{q}_e$ 在关节空间中计算。



*Positions refer to the whole robotic arm, so they are vectors.

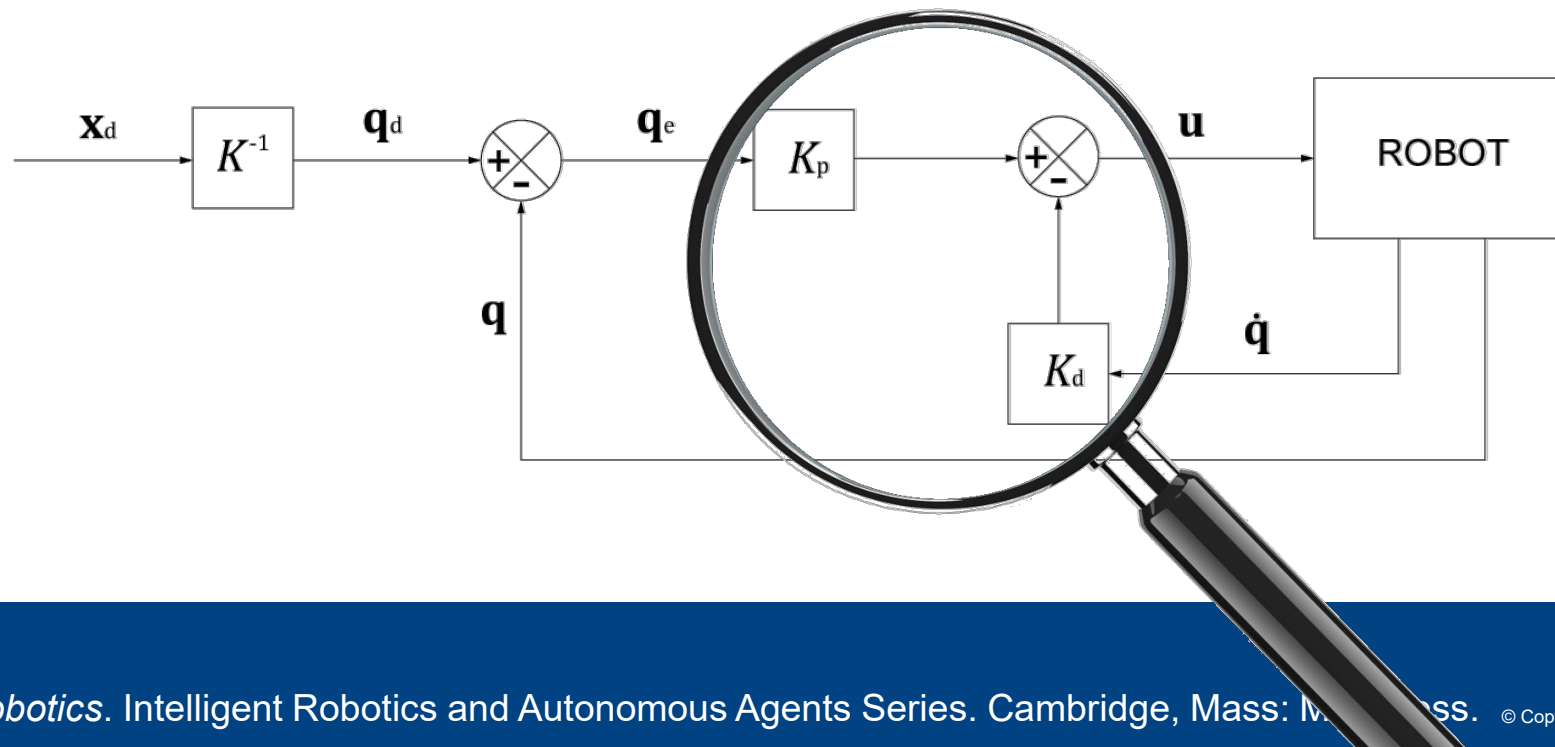
*位置指整个机械臂，因此是向量。

PD 关节空间控制

PD Joint Space Control

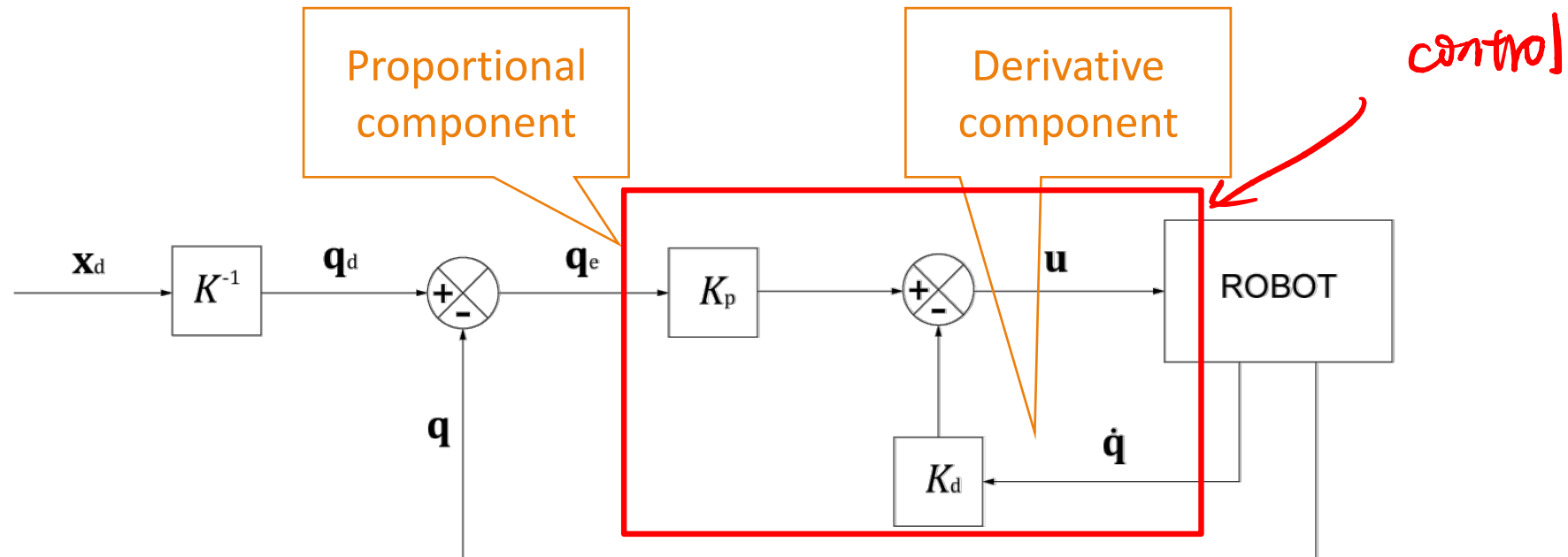
$$\mathbf{u} = \mathbf{K}_p \mathbf{q}_e + \mathbf{K}_d \dot{\mathbf{q}}$$

比例 + 微分



PD Joint Space Control

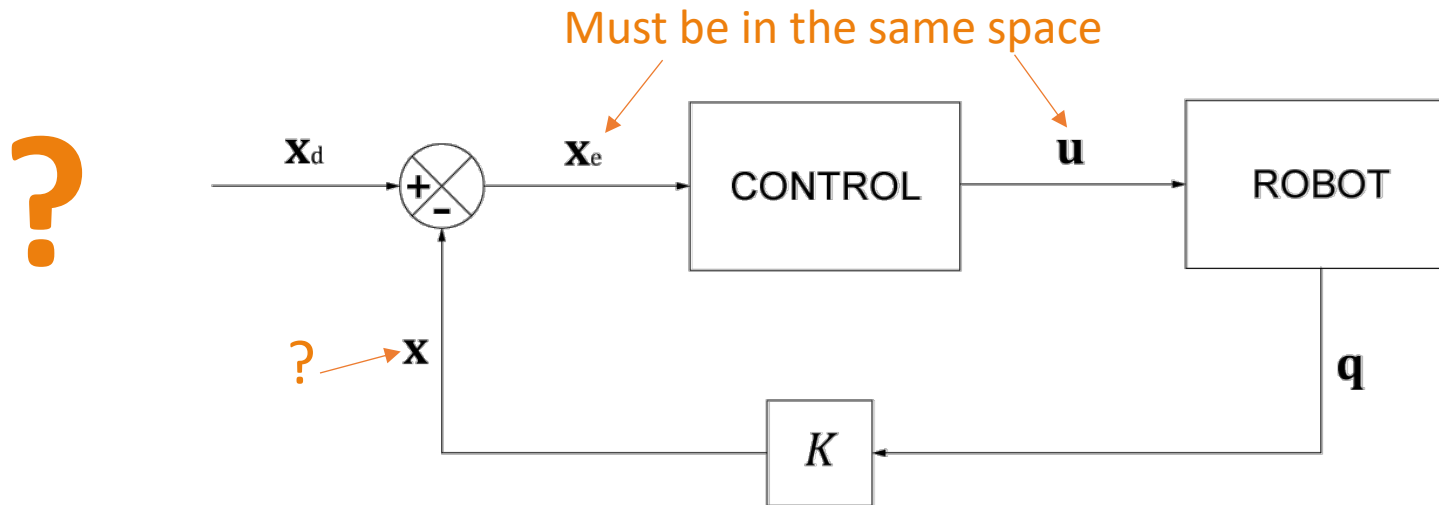
$$\mathbf{u} = \mathbf{K}_p \mathbf{q}_e + \mathbf{K}_d \dot{\mathbf{q}}$$



任务空间控制

Task Space Control

- 期望位置 $\mathbf{x}_d$ 与当前位置 $\mathbf{x}$ 直接比较，二者均在任务空间中。
- $\mathbf{x}$ 由位置传感器读数 $\mathbf{q}$ 经正运动学变换得到。
- 误差 $\mathbf{x}_e$ 在任务空间中计算。
- The desired position $\mathbf{x}_d$ is compared directly with the current position $\mathbf{x}$, both in task space.
- $\mathbf{x}$ is obtained by transforming the readings of the position sensors $\mathbf{q}$ with a forward kinematic transformation
- The error $\mathbf{x}_e$ is in task space.



正运动学：

$$\mathbf{x} = \mathbf{K}(\mathbf{q})$$

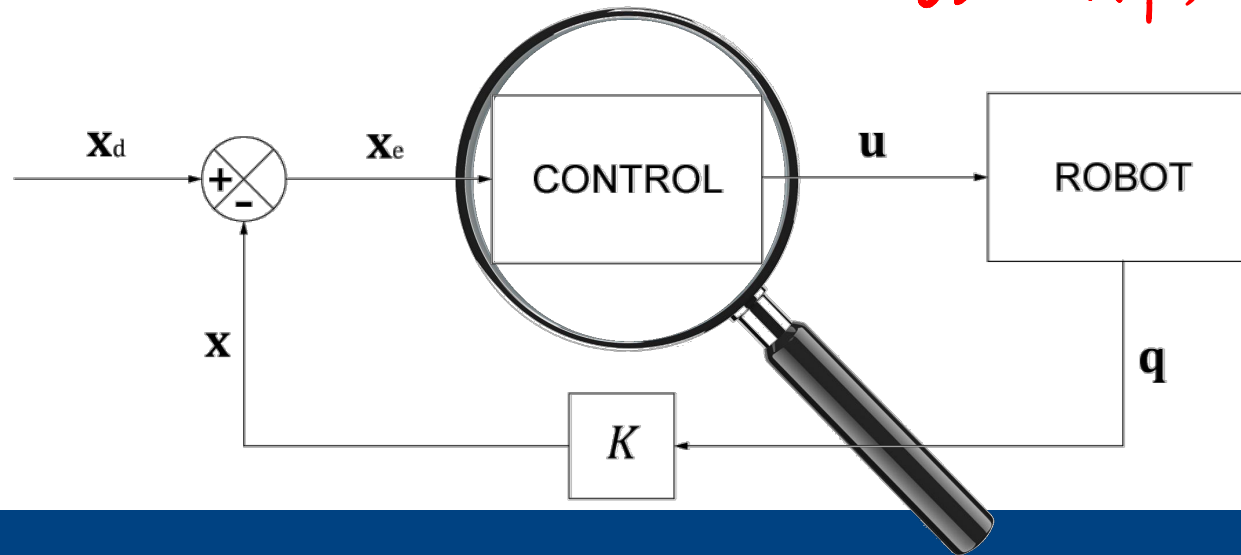
$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q}) \cdot \dot{\mathbf{q}}$$

$$\ddot{\mathbf{x}} = \mathbf{J}(\mathbf{q}) \cdot \ddot{\mathbf{q}} + \dot{\mathbf{J}}(\mathbf{q}, \dot{\mathbf{q}}) \cdot \dot{\mathbf{q}}$$

Task Space Control

- The desired position $\mathbf{x}_d$ is compared directly with the current position $\mathbf{x}$, both in task space.
- $\mathbf{x}$ is obtained by transforming the readings of the position sensors $\mathbf{q}$ with a forward kinematic transformation
- The error $\mathbf{x}_e$ is in task space.

已知 关节, 求末端



Control in Task Space Based on Inverse Jacobian

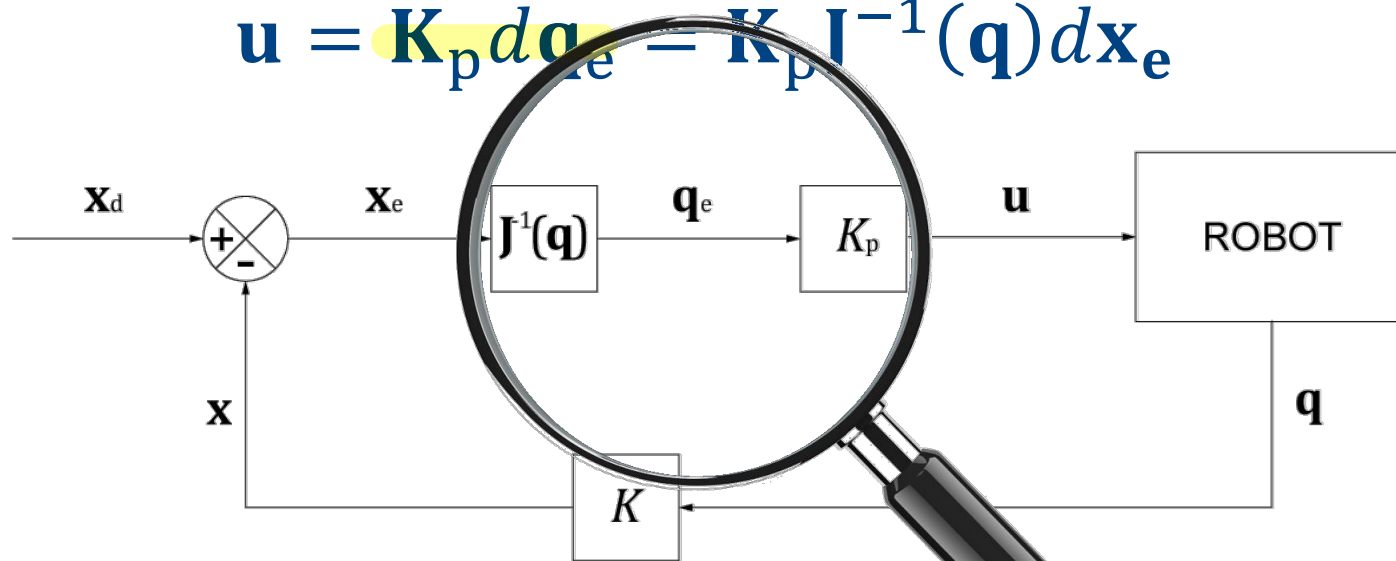
已知末端, 求关节

- 已知 $\dot{q} = J_q^{-1} \cdot \dot{x}$, 因此可以写成 $dq/dt = J_q^{-1} \cdot dx/dt$ 。
- 对于微小位移: $dq = J_q^{-1} \cdot dx$ 。
- 任务空间的误差被转换为关节空间的误差。
 $u = K_p \cdot dq_e = K_p \cdot J_q^{-1} \cdot dx_e$

- We know that $\dot{q} = J^{-1}(q)\dot{x}$, so we can write $\frac{dq}{dt} = J^{-1}(q) \frac{dx}{dt}$.
- For small displacements, $dq = J^{-1}(q)dx$.
- The error in task space is transformed into an error in joint space.

$$u = K_p dq_e = K_p J^{-1}(q) dx_e$$

末端运动误差
差换算成关节运
动误差



基于转置雅可比矩阵的任务空间运动控制

Motion control in task space based on the transposed Jacobian matrix

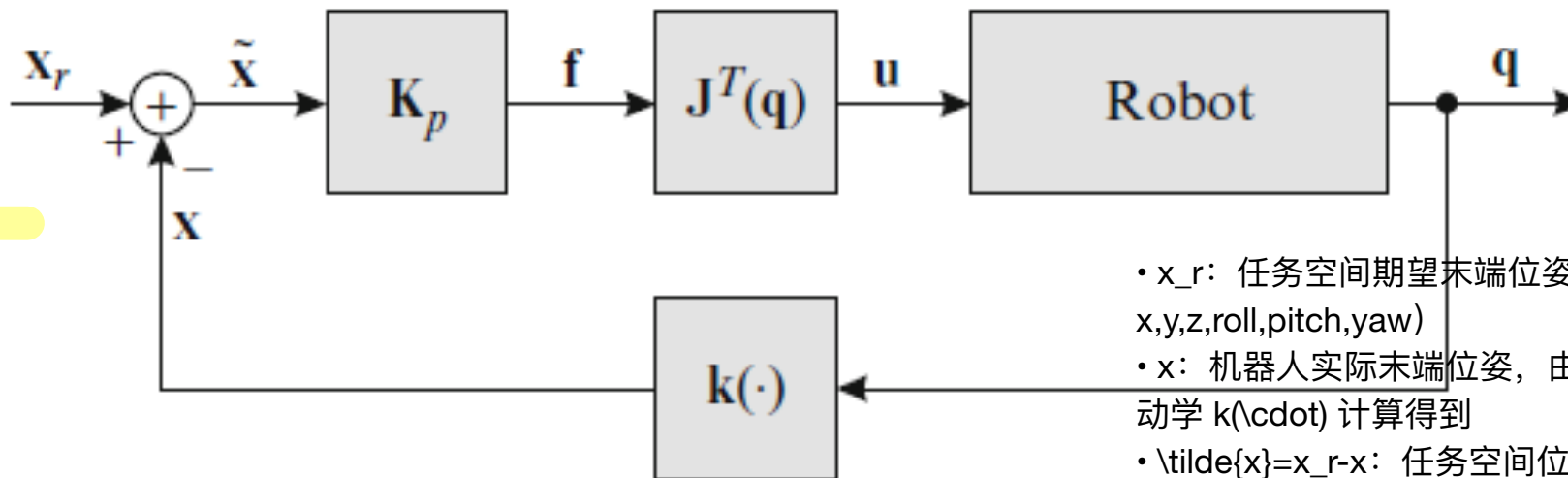
在笛卡尔空间（任务空间）算位置误差，误差乘以比例增益得到末端虚拟力；再用雅可比转置，把末端力转为关节力矩，驱动机器人向目标位置移动。

另一种转换思路：

- 直接在笛卡尔空间做位置控制，不需要求雅可比逆矩阵（规避奇异点求逆困难）

- 本质是虚拟刚度控制，把位置误差等效成弹簧力
- ✗ 局限：只有P控制，没有动力学补偿，高速运动时误差大。

Control based on the transposed Jacobian matrix



- x_r : 任务空间期望末端位姿（笛卡尔坐标系， $x, y, z, \text{roll}, \text{pitch}, \text{yaw}$ ）
- x : 机器人实际末端位姿，由关节角度 q 通过正运动学 $k(\cdot)$ 计算得到
- $\tilde{x} = x_r - x$: 任务空间位置误差
- K_p : 比例增益矩阵，把位置误差转换成末端执行器虚拟力 f
- $J^T(q)$: 雅可比转置矩阵：把笛卡尔空间的末端力映射成关节力矩 u （力的映射关系）
- Robot: 机器人本体，输入关节力矩 u ，输出关节角度 q

末端误差 → 想象末端力
→ 关节力矩。

$$\mathbf{f} = \mathbf{K}_p \tilde{\mathbf{x}}$$

$$\mathbf{u} = \mathbf{J}^T(\mathbf{q}) \mathbf{f}$$

$\mathbf{f}$ = force at end effector

• $\mathbf{f}$ = 末端执行器处的力

机器人动力学模型 (根据以上信息建模)

Robot Dynamic Model

电机力矩 τ .

(非理想积分器)

- 最终输出给执行器的力矩 τ 由以下机器人动力学模型给出:
 $\tau = B(q) \ddot{q} + C(q, \dot{q}) \dot{q} + F \dot{q} + g(q)$
- B 为惯性矩阵, 取决于当前构型 q , 乘以关节加速度 $\ddot{q}$.
- C 为科氏力矩阵, 同时取决于当前关节构型 q 和关节速度 $\dot{q}$, 乘以关节速度 $\dot{q}$.
- F 为摩擦矩阵, 乘以关节速度 $\dot{q}$.
- g 为重力项, 取决于当前机器人位置。

- The torques τ to ultimately provide to the actuators are given by the following dynamic model of the robot:

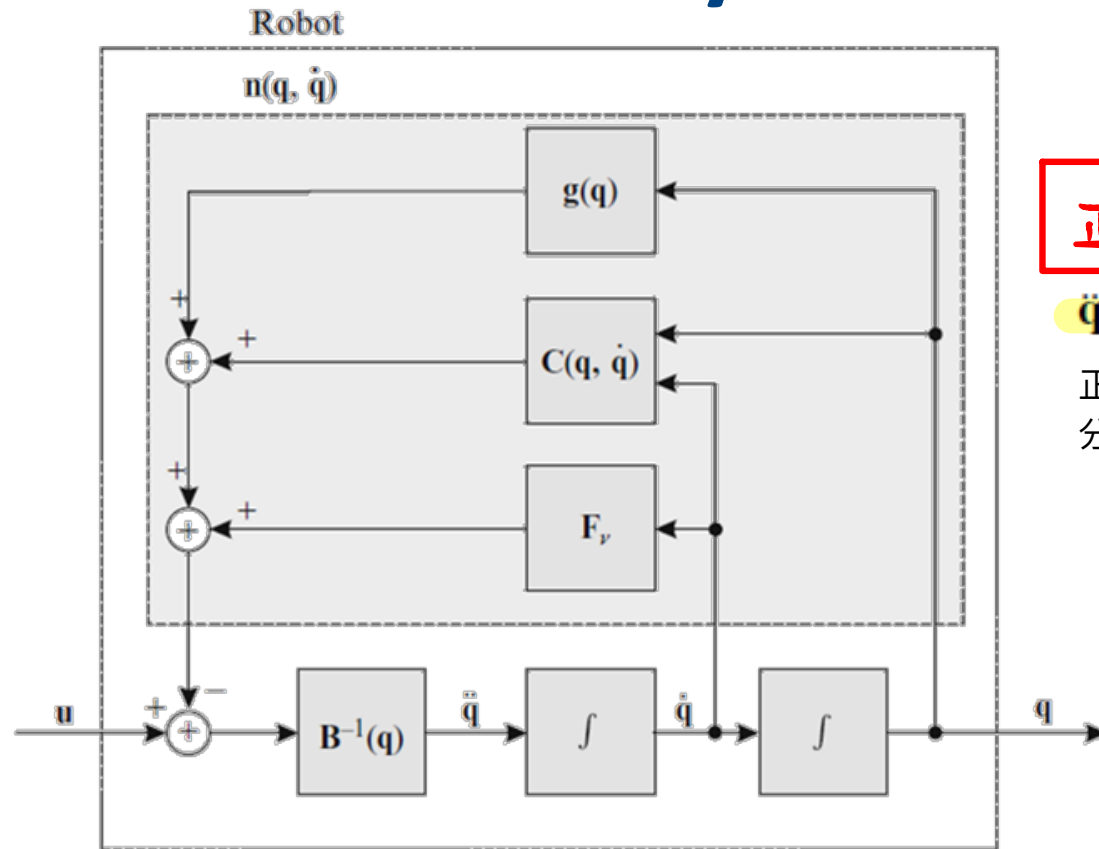
真实动力学原始公式

$$\tau = B(q) \ddot{q} + C(q, \dot{q}) \dot{q} + F \dot{q} + g(q) \rightarrow n(q, \dot{q})$$

- B is the inertia matrix, depending on current configuration q and multiplied by the joint accelerations $\ddot{q}$
- C is a matrix accounting for Coriolis forces that depend both on the current joint configuration q and on the joint velocities $\dot{q}$ and is multiplied by the joint velocities $\dot{q}$
- F is the matrix of frictions and multiplies the joint velocities $\dot{q}$
- g accounts for gravity and depends on current robot position

基于逆动力学的关节空间运动控制

Motion control in joint space based on inverse dynamics



The direct dynamic model of a robot mechanism
Robot motion under the influence of given joint torques

• u : 关节力矩输入

$$B(q)\ddot{q} + C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q) = u.$$

正动力学 ① 已知力矩, 求关节.

$$\ddot{q} = B^{-1}(q)(u - (C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q))).$$

正动力学: 给定关节力矩 u , 求解关节加速度 $\ddot{q}$; 两次积分得到关节速度 $\dot{q}$ 、关节位置 q 。

$$n(q, \dot{q}) = C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q).$$

n 代表所有非线性耦合项 (科氏、摩擦、重力)。

$$B(q)\ddot{q} + n(q, \dot{q}) = \tau.$$

②

$$\ddot{q} = B^{-1}(q)(u - n(q, \dot{q})).$$

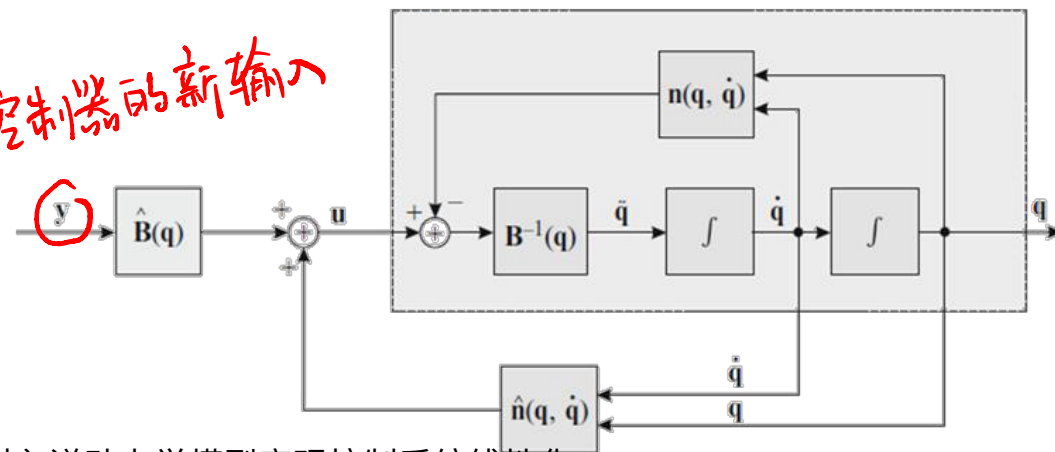
基于逆动力学的关节空间运动控制

利用逆动力学实现控制系统反
馈线性化。

Motion control in joint space based on inverse dynamics

逆动力学

控制器的新输入



$$B(q)\ddot{q} + C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q) = u.$$

已知关节, 求力矩(电机)

$$\ddot{q} = B^{-1}(q)(u - (C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q))).$$

7.7 通过引入逆动力学模型实现控制系统线性化

7.7 Linearization of the control system by implementing the inverse dynamic model

Let us assume that the robot dynamic model is known. The inertial matrix $\hat{B}(q)$ is an approximation of the real values $B(q)$, while $\hat{n}(q, \dot{q})$ represents an approximation of $n(q, \dot{q})$ as follows

$$\hat{n}(q, \dot{q}) = \hat{C}(q, \dot{q})\dot{q} + \hat{F}_v\dot{q} + \hat{g}(q). \quad \text{非线性项的估计}$$

The controller output u is determined by the following equation

控制器输出: $u = \hat{B}(q)y + \hat{n}(q, \dot{q}),$

惯性矩阵估计值

where the approximate inverse dynamic model of the robot was used.

$$n(q, \dot{q}) = C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q).$$

$$B(q)\ddot{q} + n(q, \dot{q}) = \tau.$$

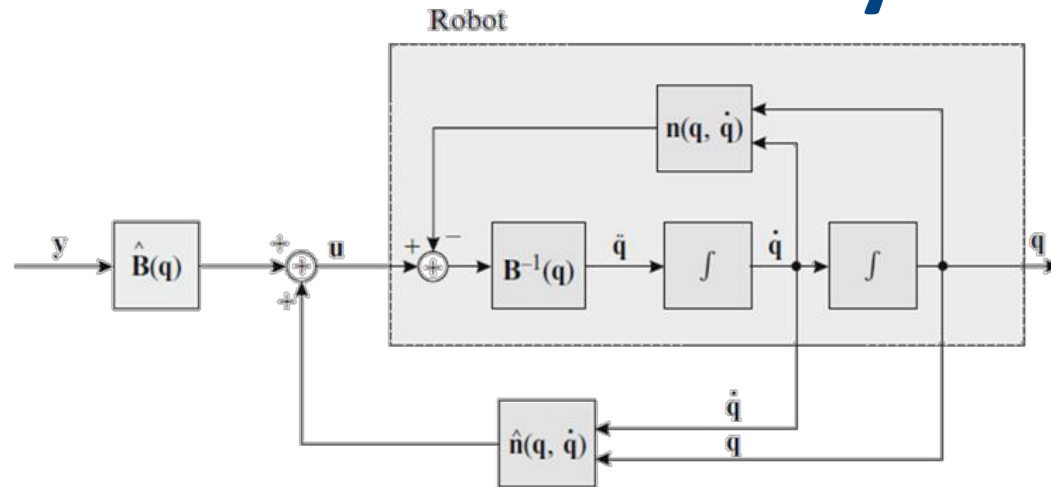
$$\ddot{q} = B^{-1}(q)(u - n(q, \dot{q})).$$

Motion control in joint space based on inverse dynamics

核心思想【反馈线性化】

控制器输出力矩分为两部分：

1. $\hat{n}$: 抵消掉机器人的重力、科氏力、摩擦非线性项
2. $B(q)y$: 抵消惯性矩阵，把原本非线性耦合机器人，等效解耦成一组独立的双积分器



7.7 Linearization of the control system by implementing the inverse dynamic model

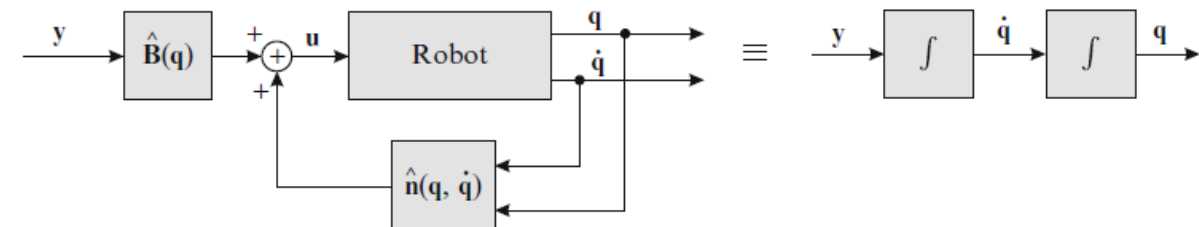
Let us assume that the robot dynamic model is known. The inertial matrix $\hat{B}(q)$ is an approximation of the real values $B(q)$, while $\hat{n}(q, \dot{q})$ represents an approximation of $n(q, \dot{q})$ as follows

$$\hat{n}(q, \dot{q}) = \hat{C}(q, \dot{q})\dot{q} + \hat{F}_v\dot{q} + \hat{g}(q).$$

The controller output u is determined by the following equation

$$u = \hat{B}(q)y + \hat{n}(q, \dot{q}),$$

where the approximate inverse dynamic model of the robot was used.



7.8 The linearized system

如果模型估计完全准确，代入动力学方程化简后，就得到：

$$\ddot{q} = y$$

也就是把原本非线性耦合的机器人，通过前馈补偿，改造为简单线性系统。这个技术叫做反馈线性化。 y 变成我们新的控制输入，我们可以自由设计 y 的控制律。

按照 u 求解结果的格式设一个 $\hat{n}$ 估计值，代入到正动力学中，

$$\ddot{q} = B^{-1}(\hat{B}y + \hat{n} - n), \text{ 若估计准了, } \ddot{q} = y$$

Motion control in joint space based on inverse dynamics

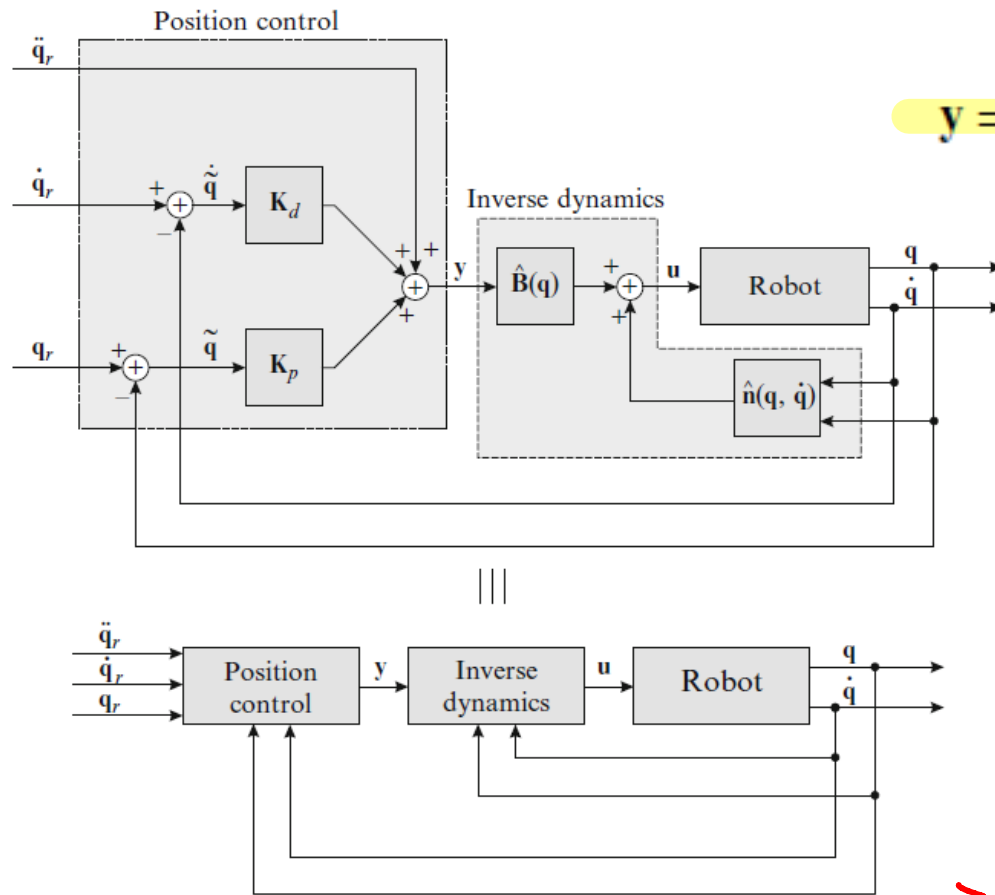


Fig. 7.9 Control of the robot based on inverse dynamics

期望 实际

$$y = \ddot{q}_r + K_p(\underline{q}_r - \underline{q}) + K_d(\dot{q}_r - \dot{q})$$

inverse dynamics 控制器里的
期望真实机器人产生的"修正后的
关节加速度"

已知(期望) 一条关节轨迹: $q_r, \dot{q}_r, \ddot{q}_r$

实际、测量: $q, \dot{q}$

$$\Rightarrow \ddot{q} = y \Rightarrow \ddot{q} = \ddot{q}_r + k_p(q_r - q) + k_d(\dot{q}_r - \dot{q})$$

Motion control in task space based on inverse dynamics

可化简为 $\ddot{\tilde{q}} + K_d \dot{\tilde{q}} + K_p \tilde{q} = 0$
(二阶线性系统)

将电机 $q \rightarrow$ 末端位移

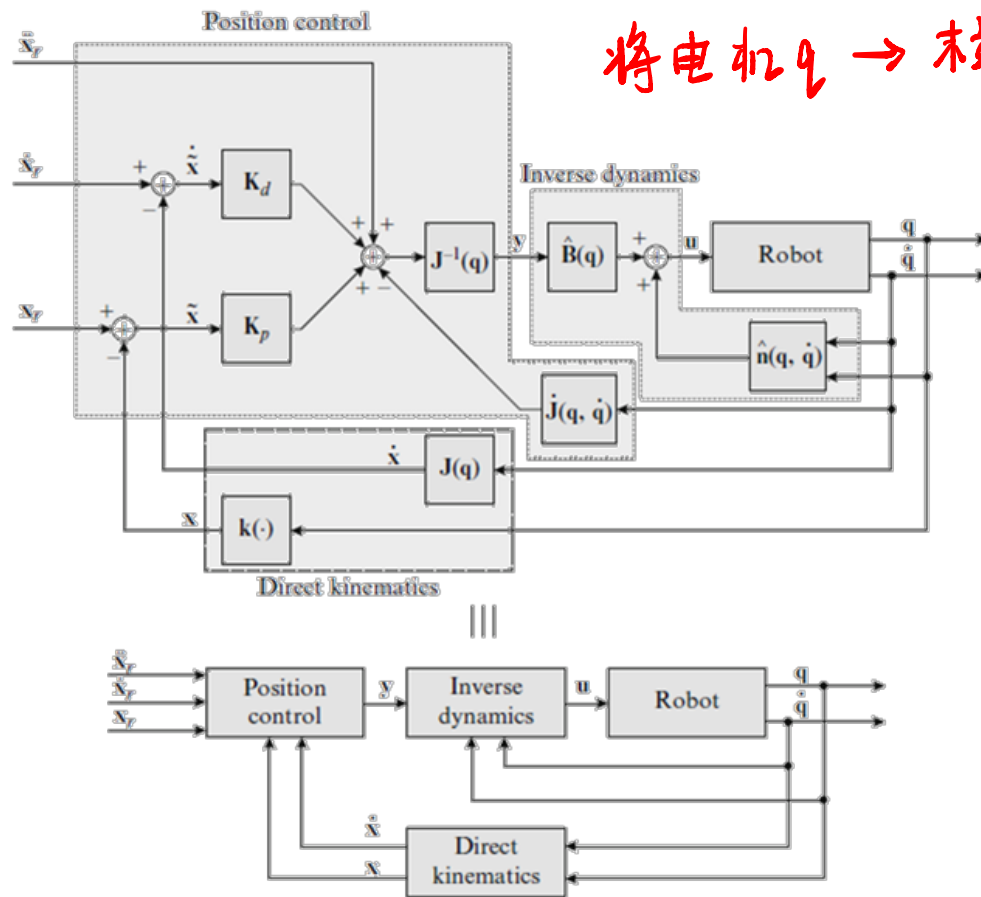


Fig. 7.13 Robot control based on inverse dynamics in external coordinates

$$\dot{x} = J(q)\dot{q}. \quad (7.33)$$

By calculating the time derivative of equation (7.33), we obtain

$$\ddot{x} = J(q)\ddot{q} + \dot{J}(q, \dot{q})\dot{q}. \quad (7.34)$$

The error of the pose of the robot end-effector is determined as the difference between its desired and its actual pose

$$\tilde{x} = x_r - x = x_r - k(q). \quad (7.35)$$

In a similar way the velocity error of the robot end-effector is determined

$$\dot{\tilde{x}} = \dot{x}_r - \dot{x} = \dot{x}_r - J(q)\dot{q}. \quad (7.36)$$

The acceleration error is the difference between the desired and the actual acceleration

$$\ddot{\tilde{x}} = \ddot{x}_r - \ddot{x}. \quad (7.37)$$

When developing the inverse dynamics based controller in the internal coordinates, equation (7.19) was derived describing the dynamics of the control error in the form $\ddot{\tilde{q}} + K_d \dot{\tilde{q}} + K_p \tilde{q} = 0$. An analogous equation can be written for the error of the end-effector pose. From this equation the acceleration $\ddot{\tilde{x}}$ of the robot end-effector can be expressed

$$\ddot{\tilde{x}} + K_d \dot{\tilde{x}} + K_p \tilde{x} = 0 \Rightarrow \ddot{\tilde{x}} = \ddot{x}_r + K_d \dot{\tilde{x}} + K_p \tilde{x}. \quad (7.38)$$

From equation (7.34) we express $\ddot{q}$ taking into account the equality $y = \ddot{q}$

$$y = J^{-1}(q) (\ddot{\tilde{x}} - \dot{J}(q, \dot{q})\dot{q}). \quad (7.39)$$

By replacing $\ddot{\tilde{x}}$ in equation (7.39) with expression (7.38), the control algorithm based on inverse dynamics in the external coordinates is obtained

$$y = J^{-1}(q) (\ddot{x}_r + K_d \dot{\tilde{x}} + K_p \tilde{x} - \dot{J}(q, \dot{q})\dot{q}). \quad (7.40)$$

The control scheme encompassing the linearization of the system based on inverse dynamics (7.31) and the closed loop control (7.40) is shown in Figure 7.13.

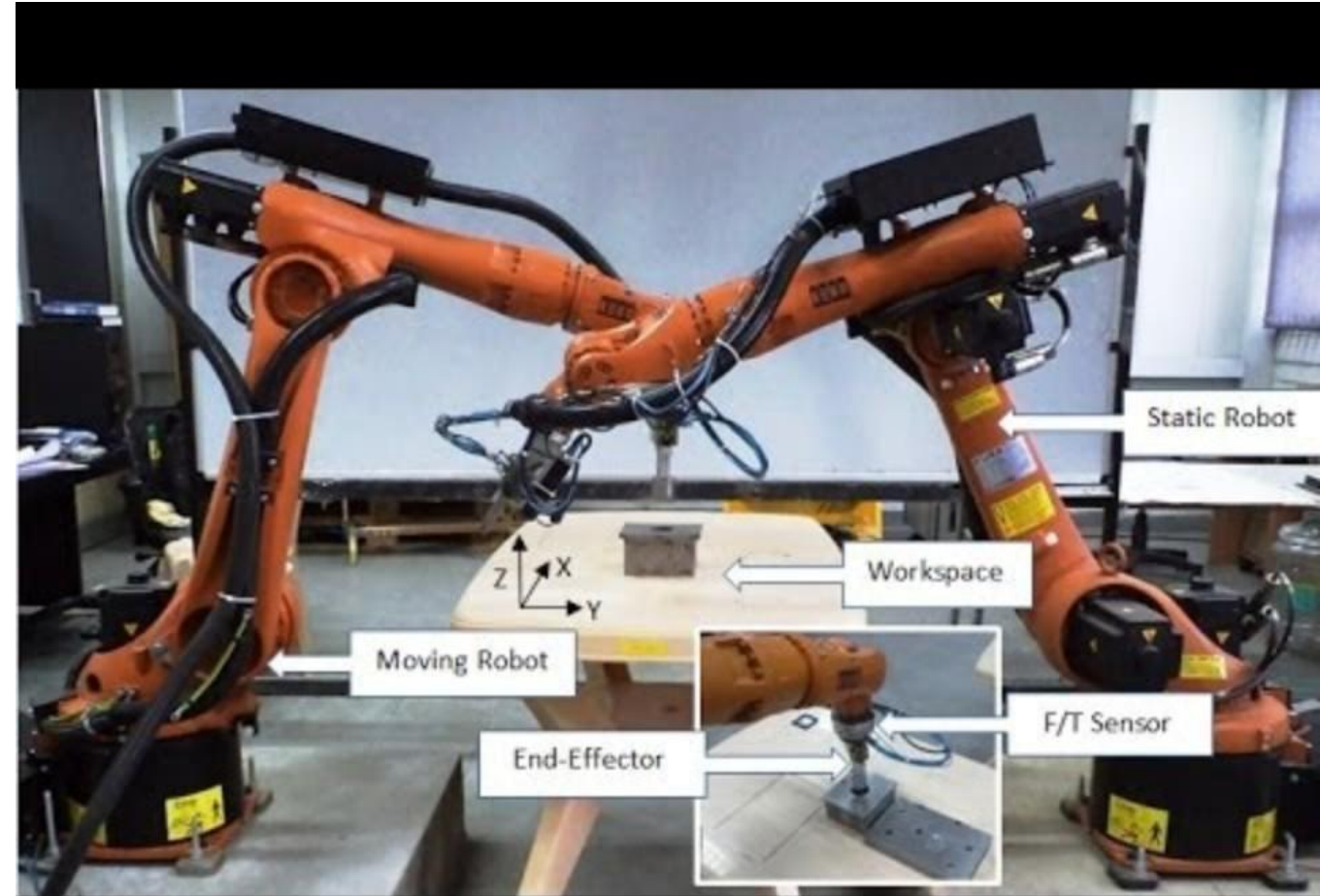
$$u = \hat{B} \cdot y + \hat{n}$$

交互控制

Interaction Control

- **Passive interaction control:** The trajectory of the end-effector is driven by the interaction forces due to the inherent nature or compliance of the robot (i.e., internally, such as joints, servo, joints, etc.).
- In passive control, the end-effector's motion naturally deflects under force, as in soft robots.
- But, this lacks flexibility (every specific task might require a special end-effector to be designed and it can also have position and orientation deviations) and high contact forces could occur because there is no force measurement.

- 被动交互控制：末端执行器的轨迹由机器人固有特性或柔顺性产生的交互力驱动（即内部层面，如关节、伺服机构等）。
- 在被动控制中，末端执行器在力作用下自然偏转，正如软体机器人那样。
- 但这种方式缺乏灵活性（每个特定任务可能需要专门设计末端执行器，且可能存在位置和姿态偏差），而且由于没有力测量，可能产生很大的接触力。



- 主动交互控制：依靠传感器（如力/力矩传感器）和/或反馈控制器来测量交互力，并相应调整机器人指令——无论是修改轨迹还是改变操作物体的方式。

Interaction Control

- **Active interaction control:** It relies on sensors (e.g., force/torque sensors) and/or feedback controllers to measure interaction forces and adjust the robot's commands accordingly—whether by modifying its trajectory or the way it manipulates objects.
- This approach enables real-time reactions to contact, offering high flexibility and accuracy.
- However, it comes with added complexity and limitations in speed.
- To achieve effective task execution and robust disturbance rejection, active control is typically combined with some degree of passive compliance. Active strategies can be further divided into
 - indirect methods (such as admittance and impedance control)
 - direct force control techniques (such as hybrid force/motion control).

- 这种方法可以对接触做出实时反应，具有很高的灵活性和精度。
- 然而，它也带来了更高的复杂度和速度上的限制。
- 为了实现有效的任务执行和强大的抗干扰能力，主动控制通常与一定程度的被动柔顺性相结合。主动策略可进一步分为：
 - 间接方法（如导纳控制和阻抗控制）
 - 直接力控制技术（如力/位置混合控制）



Robot Dynamic Model

- The torques τ to ultimately provide to the actuators are given by the following dynamic model of the robot:

$$\tau = \mathbf{B}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{F}\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$$

- $\mathbf{B}$ is the inertia matrix, depending on current configuration $\mathbf{q}$ and multiplied by the joint accelerations $\ddot{\mathbf{q}}$
- $\mathbf{C}$ is a matrix accounting for Coriolis forces that depend both on the current joint configuration $\mathbf{q}$ and on the joint velocities $\dot{\mathbf{q}}$ and is multiplied by the joint velocities $\dot{\mathbf{q}}$
- $\mathbf{F}$ is the matrix of frictions and multiplies the joint velocities $\dot{\mathbf{q}}$
- $\mathbf{g}$ accounts for gravity and depends on current robot position

Robot Dynamic Model with External Forces

最终输出给执行器的力矩 τ 由以下机器人动力学模型给出：

- The torques τ to ultimately provide to the actuators are given by the following dynamic model of the robot:

$$\tau = \mathbf{B}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{F}\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \mathbf{J}^T(\mathbf{q})\mathbf{F}_{tip}$$

- $\mathbf{J}^T(\mathbf{q})$ is the transposed manipulator Jacobian, mapping joint velocities to end-effector velocities
- $\mathbf{F}_{tip}$ is the force and moment vector applied at the end effector (tip)
 - $\mathbf{J}^T(\mathbf{q})$ 为转置操作雅可比矩阵，将关节速度映射为末端执行器速度。
 - $\mathbf{F}_{tip}$ 为作用在末端执行器（尖端）上的力和力矩向量。

准静态条件下含外力的机器人动力学模型

Robot Dynamic Model with External Forces in Quasi-static Conditions

$$\tau = \cancel{B(\mathbf{q})\dot{\mathbf{q}}} + \cancel{C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}} + \cancel{F\dot{\mathbf{q}}} + \mathbf{g}(\mathbf{q}) + \mathbf{J}^T(\mathbf{q})\mathbf{F}_{tip}$$

- 在被动交互控制中， $\mathbf{F}_{tip}$ 由环境施加，由此产生的运动取决于机器人的固有柔顺性。

$$\tau = \mathbf{g}(\mathbf{q}) + \mathbf{J}^T(\mathbf{q})\mathbf{F}_{tip}$$

- In **passive interaction control**, $\mathbf{F}_{tip}$ is imposed by the environment, and the resulting motion depends on the robot's intrinsic compliance.
- In **active interaction control**, this equation becomes a control law: we specify a desired contact force $\mathbf{F}_d$ and compute the torque needed to apply it:

• 在主动交互控制中，该方程变为控制律：我们指定期望接触力 $\mathbf{F}_d$ ，并计算施加该力所需的力矩：

$$\tau = \tilde{\mathbf{g}}(\mathbf{q}) + \mathbf{J}^T(\mathbf{q})\mathbf{F}_d$$

with $\tilde{\mathbf{g}}(\mathbf{q})$ an estimation of gravitational torque 其中 $\hat{\mathbf{g}}(\mathbf{q})$ 为重力力矩的估计值。

Interaction Control in Quasi-static Conditions

$$\tau = \tilde{g}(\mathbf{q}) + J^T(\mathbf{q})\mathbf{F}_d$$

$$\tau = \tilde{g}(\mathbf{q}) + J^T(\mathbf{q}) \left(\mathbf{F}_d + \mathbf{K}_p \mathbf{F}_e + \mathbf{K}_i \int \mathbf{F}_e dt \right)$$

PI controller

- $\mathbf{F}_e = \mathbf{F}_d - \mathbf{F}_{tip}$ is the force error
- $\mathbf{K}_p$ is the proportional gain matrix
- $\mathbf{K}_i$ is the integral gain matrix

• $\mathbf{F}_e$ 为力误差: $\mathbf{F}_e = \mathbf{F}_d - \mathbf{F}_{tip}$

• $\mathbf{K}_p$ 为比例增益矩阵

• $\mathbf{K}_i$ 为积分增益矩阵
(比例分量 + 积分分量)

Integral
component

Proportional
component

机械阻抗 Mechanical Impedance

物理系统在受力时抵抗运动的方式。

与静态刚度（将力与位移关联）不同，阻抗捕捉的是动态行为，即速度 v （或加速度）如何影响系统产生或吸收的力 F 。

Mechanical Impedance Z :

how a physical system resists motion when subjected to a force.

Unlike static **stiffness** (which links force to displacement), impedance captures **dynamic behavior**, i.e. how velocity v (or acceleration) influences the force F a system generates or absorbs.

$$F = Zv$$

A combination of:

- **Inertial response (mass):** $F = Ma$
- **Elastic response (Hooke's law):** $F = Kx$
- **Damping behavior:** $F = -Bv$

由以下三部分组合而成：

- 惯性响应（质量）： $F = M \cdot a$
- 弹性响应（胡克定律）： $F = K \cdot x$
- 阻尼行为： $F = -B \cdot v$

阻抗控制

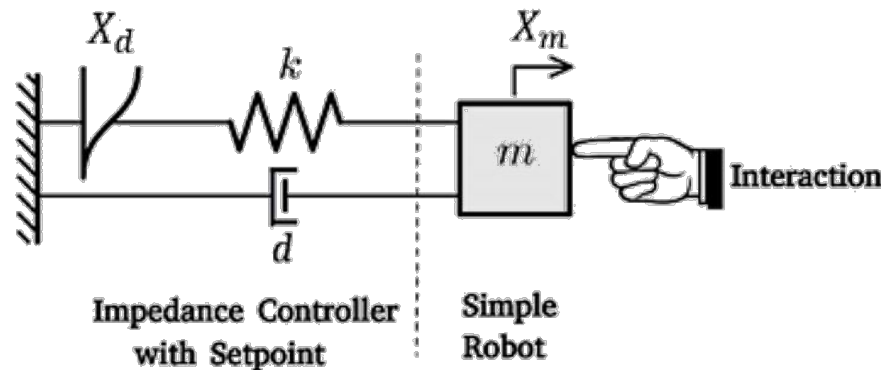
Impedance Control

$$M_d \ddot{x} + B_d \dot{x} + K_d(x - x_r) = f_{ext}$$

- x is the end-effector position
- x_r is the reference trajectory
- M_d is the desired virtual **mass** matrix
- B_d is the desired virtual **damping** matrix
- K_d is the desired virtual **stiffness** matrix
- f_{ext} is the external force exerted by the environment

- x 为末端执行器位置
- x_r 为参考轨迹
- M_d 为期望虚拟质量矩阵
- B_d 为期望虚拟阻尼矩阵
- K_d 为期望虚拟刚度矩阵
- f_{ext} 为环境施加的外力

- 高刚度 (K_d) 使机器人更刚硬, 抵抗偏离 x_r 。
- 高阻尼 (B_d) 使运动平滑, 减少振动。
- 高惯性 (M_d) 使机器人对外界突发扰动不那么敏感。



Robot as a Spring-Mass-Damper System

- A high stiffness (K_d) makes the robot stiffer, resisting deviations from x_r .
- A high damping (B_d) smooths out motion and reduces vibrations.
- A high inertia (M_d) makes the robot less sensitive to sudden external disturbances.

Stiffness Control 刚度控制（阻抗控制的子集） (A Subset of Impedance Control)

- 静态行为
 - **Static behaviour** • 我们不指定机器人的动态响应方式（质量、阻尼、刚度），而只定义位置/姿态偏差与施加到环境的力之间的静态关系。
 - Instead of specifying how the robot responds dynamically (with mass, damping, and stiffness), we only define a static relationship between the deviation in position/orientation and the force exerted on the environment. • 这通过调整阻抗模型中刚度 K 的元素来实现，同时忽略惯性项和阻尼项。
 - This is done by acting on the elements of the stiffness K in the impedance model while ignoring inertial and damping terms
 - 高刚度 (K_d) → 位置跟踪精确，但柔顺性较差
 - 低刚度 (K_d) → 柔顺性更好，但在外部力作用下允许更大的运动偏差。
- High stiffness (K_d) → accurate position tracking, but less compliant
- Low stiffness (K_d) → more compliant, but allows larger motion deviations under external forces.

导纳控制 (阻抗的对偶)

Admittance Control (Dual of Impedance)

- 在阻抗控制中，我们定义机器人应根据运动偏差施加多大的力。
- 在导纳控制中，正好相反：我们测量外力，并计算机器人应移动多少来顺应这个力。
- In impedance control, we define **how much force** the robot should apply in response to a deviation in motion.
- In admittance control, it's the opposite: we measure an external force and compute **how much the robot should move** to accommodate that force.
 - 例如，如果有10N的力将机器人推离路径，导纳控制器可能会说“退让1毫米”（取决于柔顺性设置）——实际上，机器人会稍微移动，直到力减小。
- For instance, if a force of 10N is pushing the robot off its path, an admittance controller might say “yield by 1 mm” (depending on a compliance setting) – effectively, the robot moves slightly until the force reduces.
- Admittance control often involves two loops:
 - 导纳控制通常包含两个回路：
 - an **outer force loop** that modifies the target position based on force input.
 - an **inner high-bandwidth control loop** that ensures accurate tracking of the compliant position or velocity reference.
 - 外环：力回路，根据力输入修改目标位置。
 - 内环：高带宽控制回路，确保精确跟踪柔顺位置或速度参考。

外环：虚拟机械行为

Admittance Control

Outer Loop: Virtual Mechanical Behaviour

$$M_d(\ddot{x}_c - \ddot{x}_r) + B_d(\dot{x}_c - \dot{x}_r) + K_d(x_c - x_r) = f_e$$

- x_c is the compliant position generated by the controller 由控制器生成的柔顺位置
- x_r is the desired nominal trajectory 期望名义轨迹
- M_d is the desired virtual **mass** matrix 期望虚拟质量矩阵
- B_d is the desired virtual **damping** matrix 期望虚拟阻尼矩阵
- K_d is the desired virtual **stiffness** matrix 期望虚拟刚度矩阵
- f_e is the external force measured at the end-effector 末端执行器处测量的外力

In quasi-static conditions: $\dot{x}_r = 0$ and $\ddot{x}_r = 0$, so:

准静态条件下

$$M_d\ddot{x}_c + B_d\dot{x}_c + K_d(x_c - x_r) = f_e$$

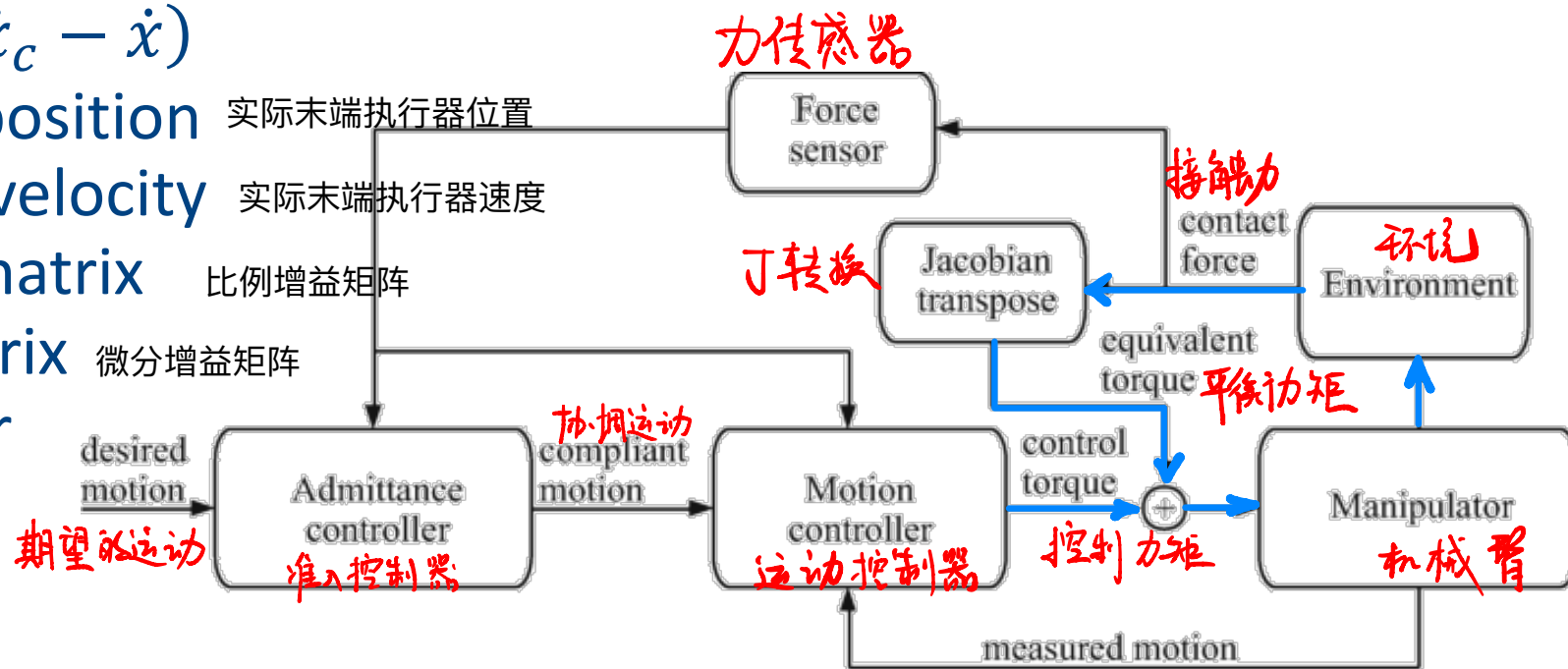
Admittance Control

内环：跟踪来顺运动

Inner Loop: Tracking the Compliant Motion

$$F = K_p(x_c - x) + K_d(\dot{x}_c - \dot{x})$$

- x : actual end-effector position 实际末端执行器位置
- $\dot{x}$: actual end-effector velocity 实际末端执行器速度
- K_p : proportional gain matrix 比例增益矩阵
- K_d : derivative gain matrix 微分增益矩阵
- F : commanded force or torque to the low-level controller 发送给低层控制器的指令力或力矩



The admittance controller computes a compliant trajectory x_c based on the external force input f_e and the desired behavior (mass-spring-damper model). The inner loop tracks x_c using a motion controller that ensures accurate execution.

工作原理:

导纳控制器根据外力输入 f_e 和期望行为（质量-弹簧-阻尼模型）计算机器柔顺轨迹 x_c 。内环通过运动控制器跟踪 x_c ，确保精确执行。

力控制 Force Control

- 利用与环境交互的显式反馈来调节机器人施加的力。
- Uses explicit feedback from the interaction with the environment to regulate the forces applied by the robot.
- Based on this contact force, the controller adjusts the robot joint torques or positions to achieve a desired force profile necessary to complete a task.
- 根据这个接触力，控制器调整机器人关节力矩或位置，以实现完成任务所需的期望力曲线。

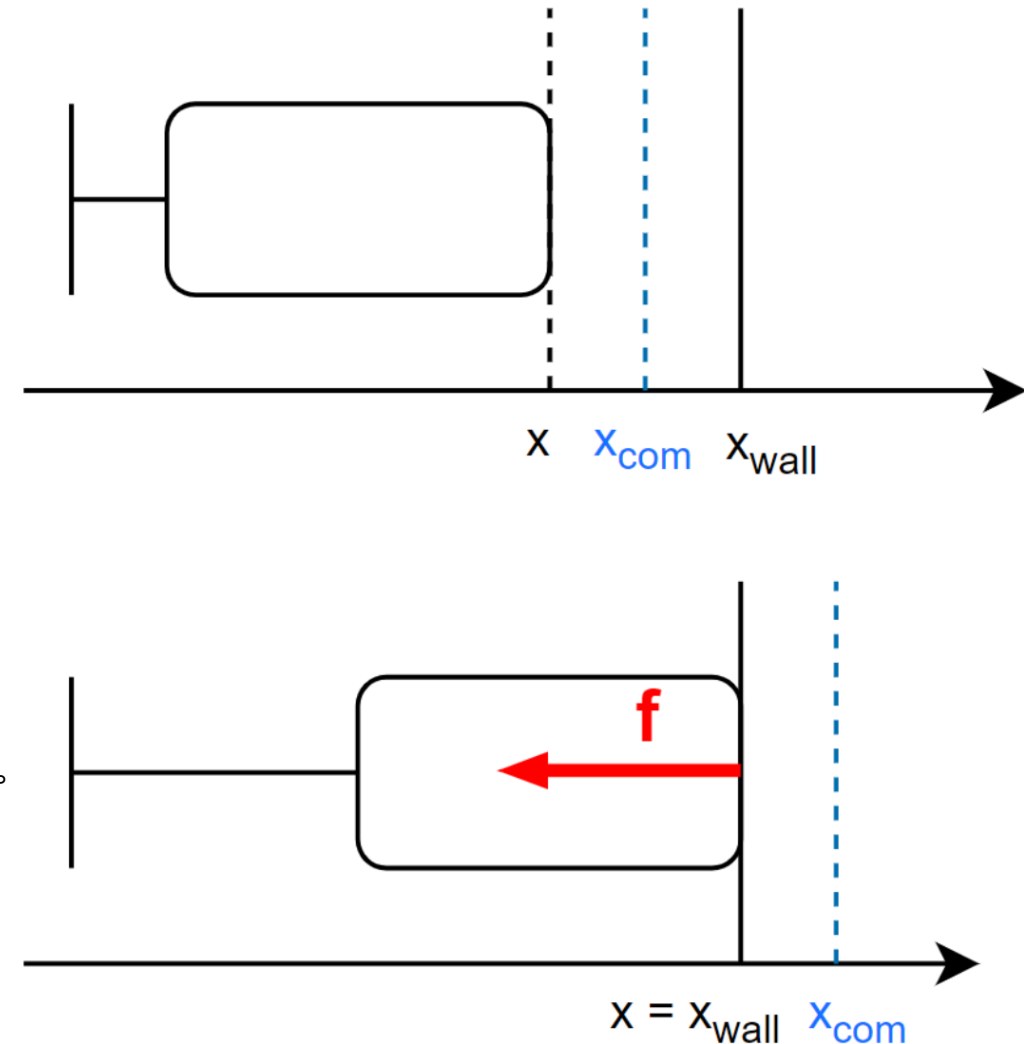
1-DOF, linear, position-controlled robot.

The goal is to control the contact force f between the end-effector and a wall so that it tracks a desired force f_d .

示例：

1自由度、线性、位置控制机器人。

目标是控制末端执行器与墙壁之间的接触力 f ，使其跟踪期望力 f_d 。

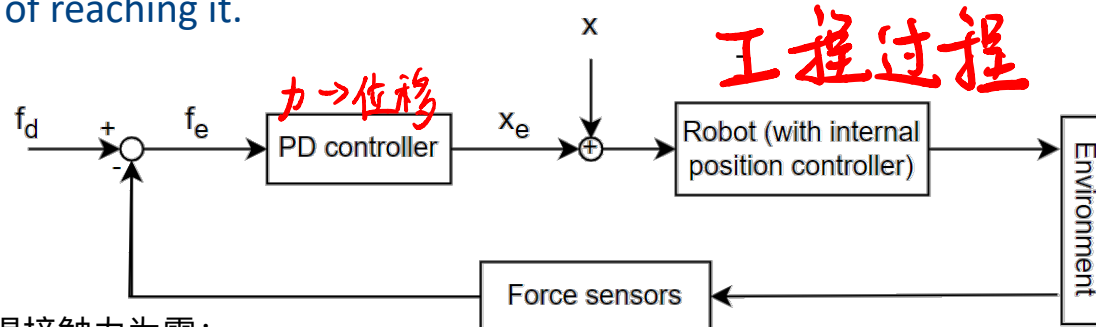


Force Control

- 比例-微分 (PD) 控制律将力误差 f_e 转换为位置误差 x_e 。

- Proportional-Derivative (PD) control law to convert the error f_e to a position error x_e .
- Note that the command sent to the robot is a computed desired position $x_{com} = x + x_e$ and its internal position controller takes care of reaching it.

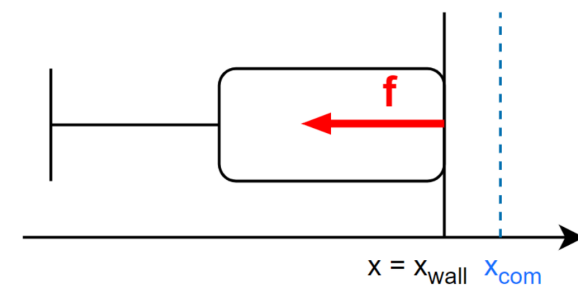
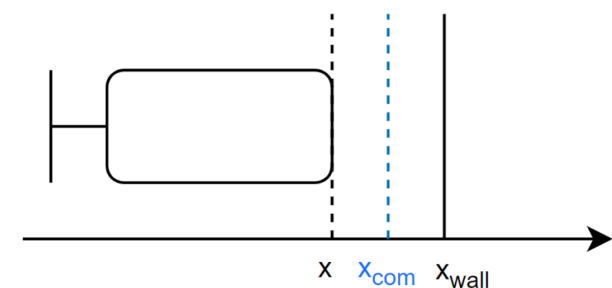
- 注意：发送给机器人的指令是计算出的期望位置 $x_{com} = x + x_e$ ，由机器人内部的位置控制器负责到达该位置。



当没有接触时，力/力矩传感器测得接触力为零：

- When there is no contact, the force/torque sensor measures zero contact force: $f = 0$, so the error $f_e = f_d - f = f_d > 0$.
- Assuming stationary initial conditions, this creates a position error $x_e = x_{com} - x > 0$, which leads to $x_{com} > 0$ which in turn makes the robot move right toward the wall. • 假设初始条件静止，这产生位置误差 $x_e = x_{com} - x > 0$ ，导致 $x_{com} > 0$ ，进而使机器人向右朝向墙壁移动。
- Upon contact between the robot end-effector and the wall, the sensed force becomes positive but still smaller than f_d assuming “soft” collision. Therefore, initially, one still has $x_{com} > x = x_{wall}$ meaning that the robot is commanded to penetrate the wall. • 当机器人末端执行器与墙壁接触后，测得的力变为正，但假设是“软”碰撞，仍小于 f_d 。因此，初始阶段仍有 $x_{com} > x = x_{wall}$ ，意味着机器人被指令穿透墙壁。
- Finally, interactions between the PD controller and the stiffness of the end-effector, of the wall and of the robot’s internal position controller (characterized in part by its own PID gains) brings the system to a steady state where:
 - $f_e = 0$
 - $x_{com} > x_{wall}$ — i.e. the robot slightly compresses the contact, holding a constant desired force.

即机器人轻微压缩接触面，保持恒定的期望力。



力/位置混合控制.

Hybrid Force/Position Control

$$B(q)\ddot{q} + C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q) = \tau - J^T(q)f.$$

末端接触力

Effect of external forces

$$n(q, \dot{q}) = C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q),$$

$$B(q)\ddot{q} + n(q, \dot{q}) = \tau - J^T(q)f.$$

$$B(q)\ddot{q} + n(q, \dot{q}) + J^T(q)f = u.$$

$$\ddot{q} = B^{-1}(q)(u - n(q, \dot{q}) - J^T(q)f).$$

$$u = \hat{B}(q)y + \hat{n}(q, \dot{q}) + J^T(q)f,$$

在原本
逆动力学的基础上,
把“接触力”也补偿掉.

若理想估计

$$\hat{B}(q) = B(q)$$

$$\hat{n} = n$$

$$\Rightarrow y = \ddot{q}$$

正动力学公式

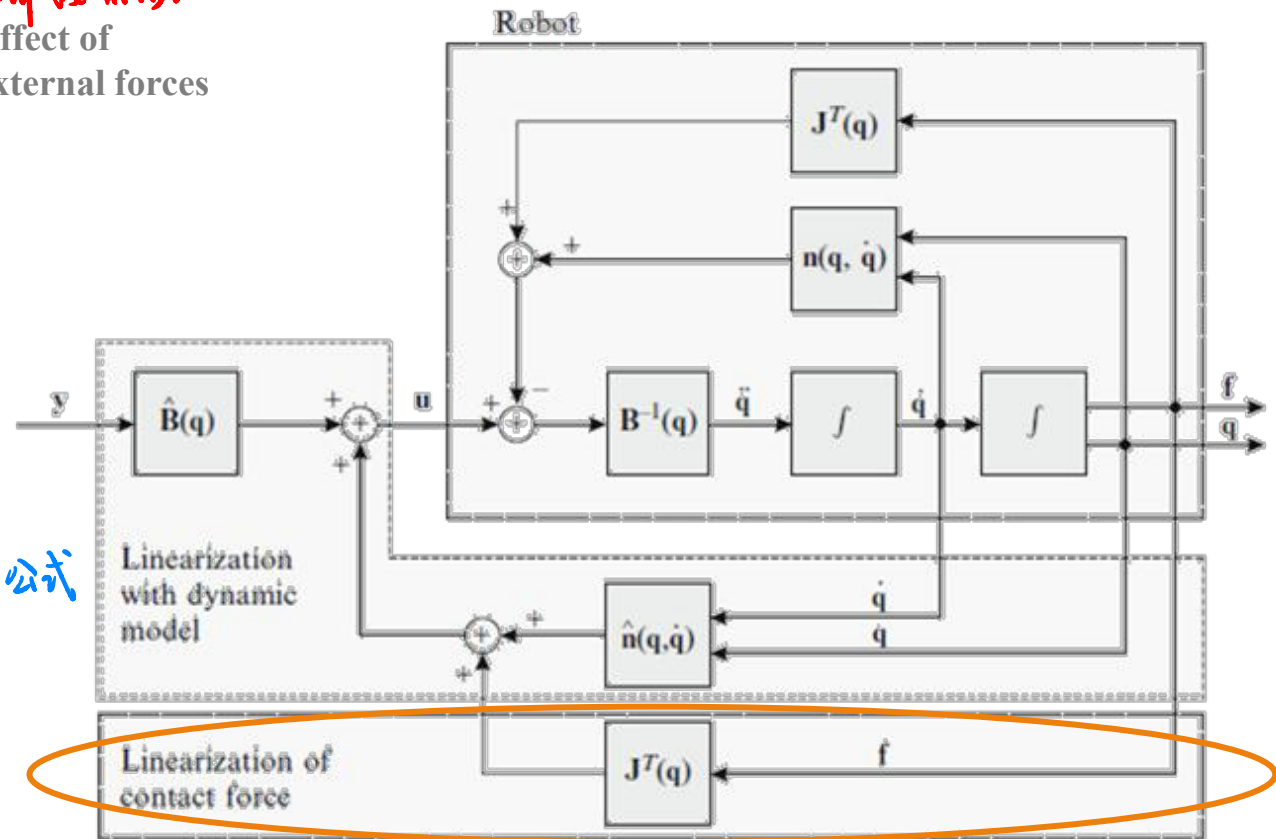


Fig. 7.14 Linearization of the control system by implementing the inverse dynamic model and the measured contact force

Hybrid Force/Position Control

在 y 之间加上

PD 把 $f \rightarrow q$ 的关系加上

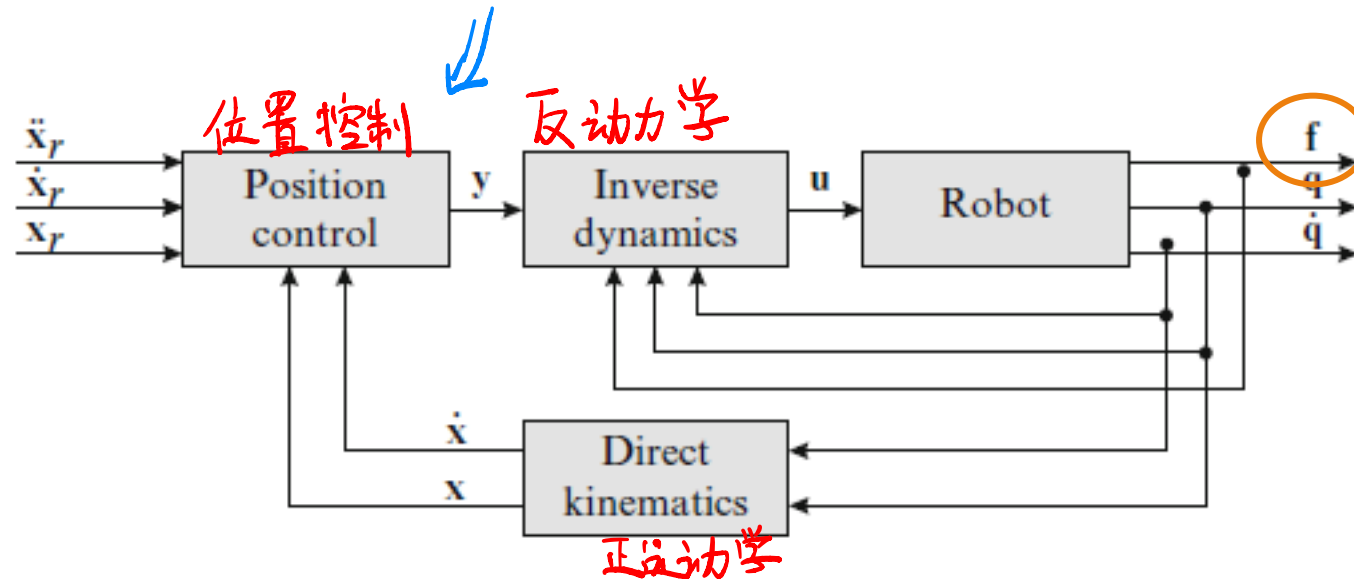
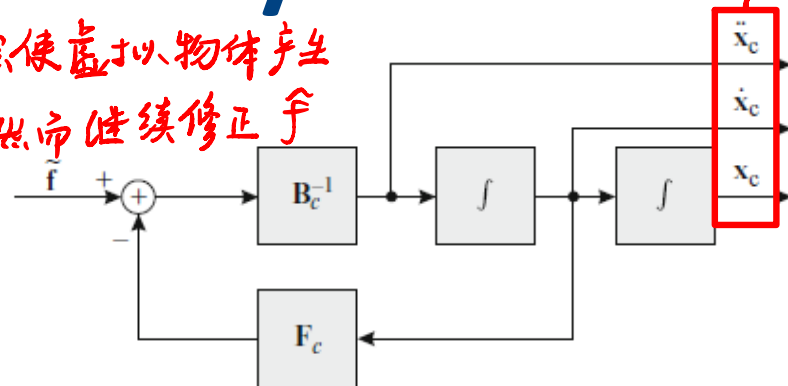


Fig. 7.15 Robot control based on inverse dynamics in external coordinates including the contact force

Hybrid Force/Position Control

有力误差 $\tilde{f}$, 就会使虚拟物体产生运动, 得到 $\tilde{x}_c$, 然而继续修正 $\tilde{f}$



由于力误差而额外产生的位置修正

$$\tilde{f} = f_d - f$$

(人工假设虚拟物体)

$$\tilde{f} = B_c \ddot{\tilde{x}}_c + F_c \dot{\tilde{x}}_c \quad (7.48)$$

B_c 和 F_c 决定了物体在手作用下的运动, 产生加速度 $\ddot{\tilde{x}}_c$.
The matrices B_c and F_c determine the movement of the object under the influence of the force $\tilde{f}$. From equation (7.48) the acceleration of the virtual object can be calculated

$$\ddot{\tilde{x}}_c = B_c^{-1} (\tilde{f} - F_c \dot{\tilde{x}}_c) \quad (7.49)$$

B_c and F_c are inertia and damping, respectively, of a virtual object.

惯性矩阵 阻尼矩阵

Fig. 7.16 Force control translated into control of the pose of robot end-effector

期望力 $\rightarrow$ 力控制
 $\rightarrow$ 修正后的末端运动 $\tilde{x}_c$, 再与原 x_d 合成,
 $\tilde{x}_c + x_d \rightarrow x_r$

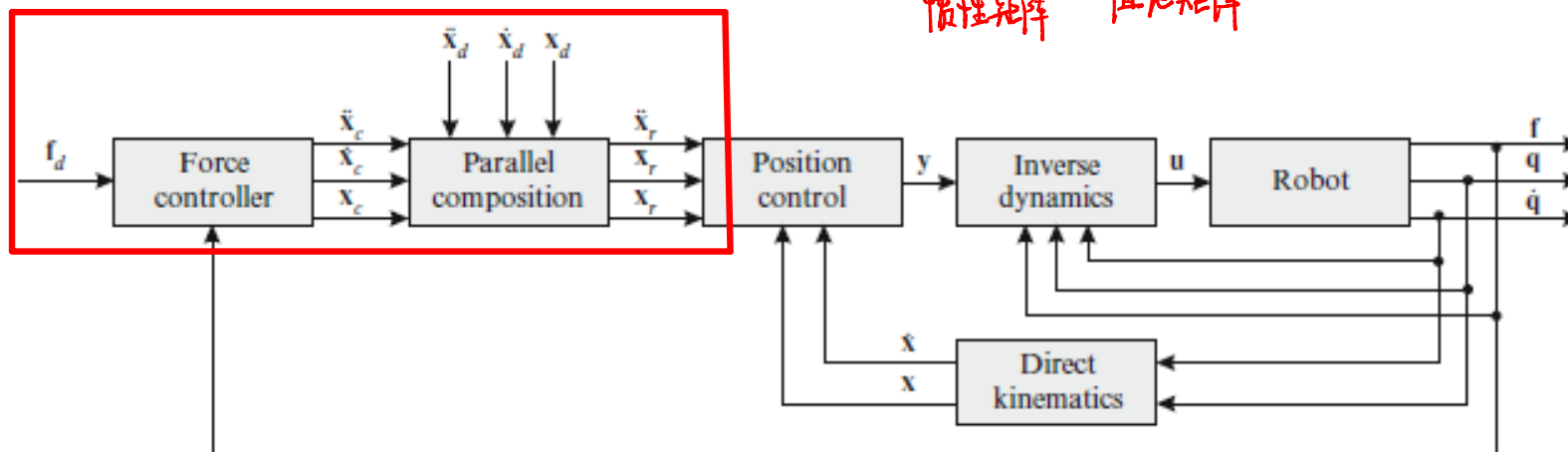


Fig. 7.17 Direct force control in the external coordinates

机器人神经控制器

Robot Neuro-Controllers

Robot Controllers

基于机器人模型.

Based on robot models

目标末端执行器位置

Target end effector
position:

$$\mathbf{x} = [x, y, z, \phi, \theta, \psi]$$

当前末端执行器位置

Current end effector
position:

$$\mathbf{x} = [x, y, z, \phi, \theta, \psi]$$

Controller

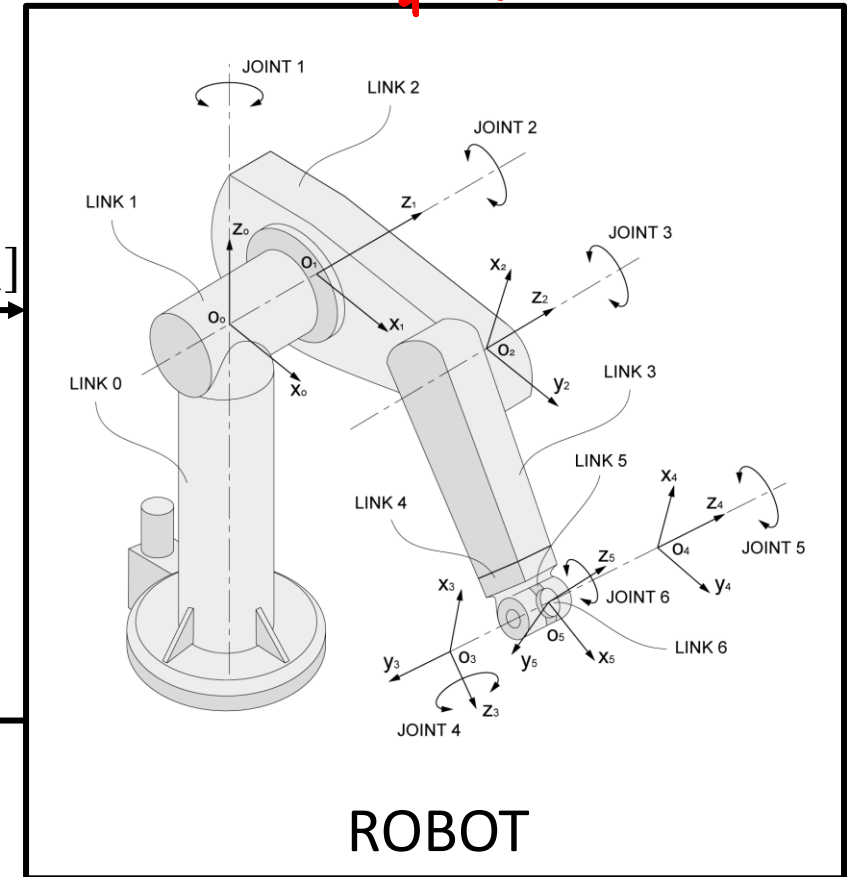
控制器.

关节驱动指令

Actuation
commands $\mathbf{u}$
to joints:

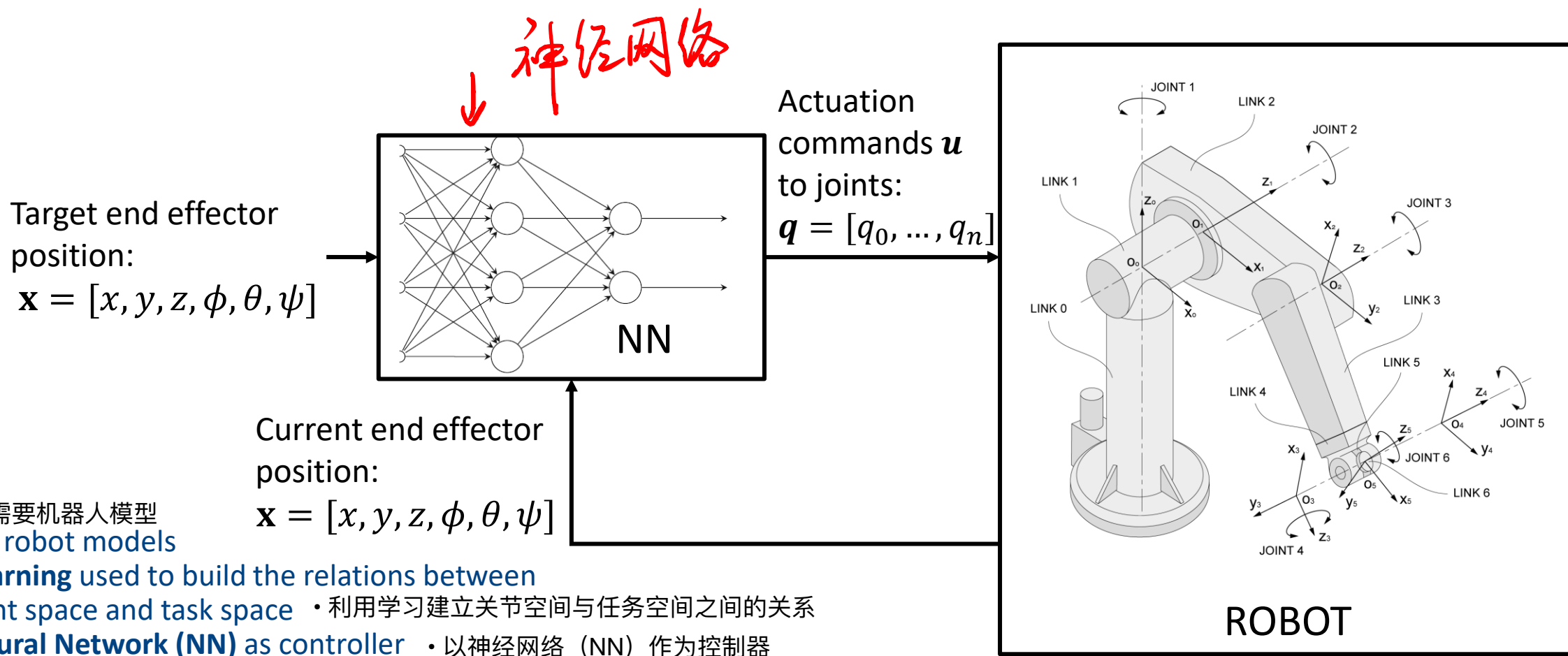
$$\mathbf{q} = [q_0, \dots, q_n]$$

机器人



机器人神经控制器.

Robot Neurocontrollers



- 不需要机器人模型
- No robot models
- Learning used to build the relations between joint space and task space • 利用学习建立关节空间与任务空间之间的关系
- Neural Network (NN) as controller • 以神经网络 (NN) 作为控制器

淋浴任务控制

Showering Task Control

NATIONAL RESEARCH FOUNDATION
PRIME MINISTER'S OFFICE
SINGAPORE

CARTIN:

Centre for Advanced Robotics
Technology Innovation
Medium Sized Centre (MSC)

控制盒

Control
Box

计算机

Computer

人体模型

Dummy
Body

气泵

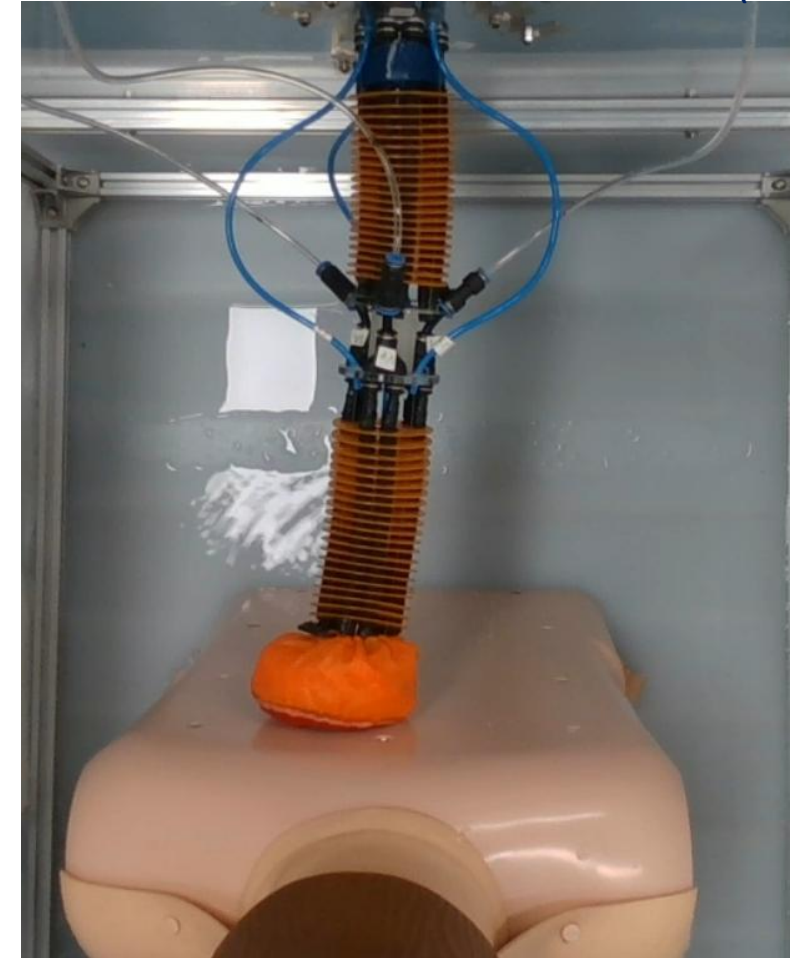
Air Pump

Sponge

海绵球

Soft Arm

柔性手臂



非传统执行器

Non-conventional actuators

Elastic actuators

弹性执行器 Elastic Actuators

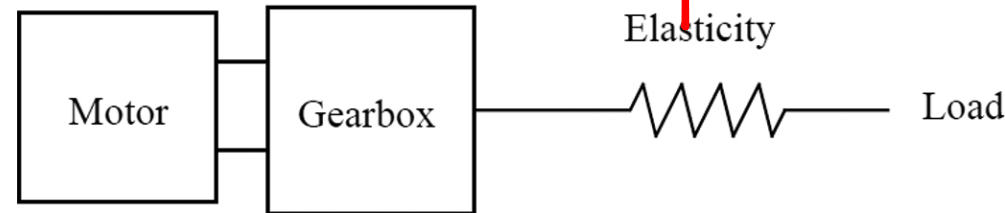
- 工业机器人（结构化环境）：执行器应尽可能刚硬，以提高精度和可控性。
- 服务机器人（非结构化环境）：柔顺性可以提供适应性。
- **Industrial robotics** (structured environment): actuators to be as stiff as possible, to increase accuracy and controllability
- **Service robotics** (unstructured environment): compliance can provide adaptability
- An elastic actuator contains an **elastic element**:
 - tolerance to impact loads,
 - low mechanical output impedance,
 - passive mechanical energy storage,
 - increased peak power output.

- 弹性执行器包含一个弹性元件：
 - 对冲击载荷的容忍度
 - 低机械输出阻抗
 - 被动机械储能
 - 提高峰值功率输出

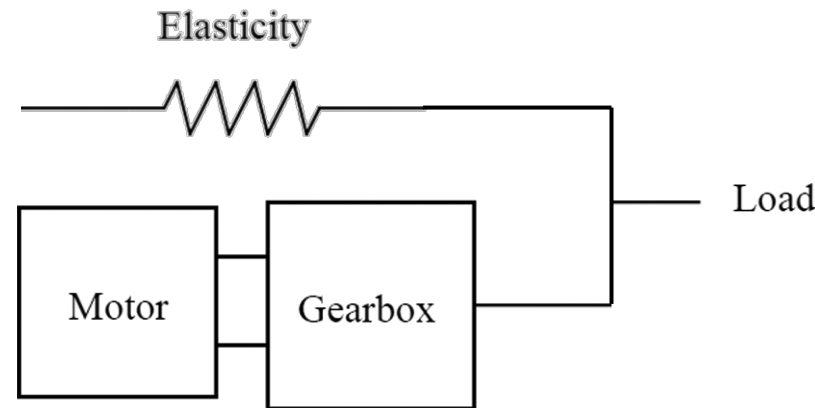
弹性执行器 一 布置方式

Elastic Actuators – Arrangement

- Serial 串联式



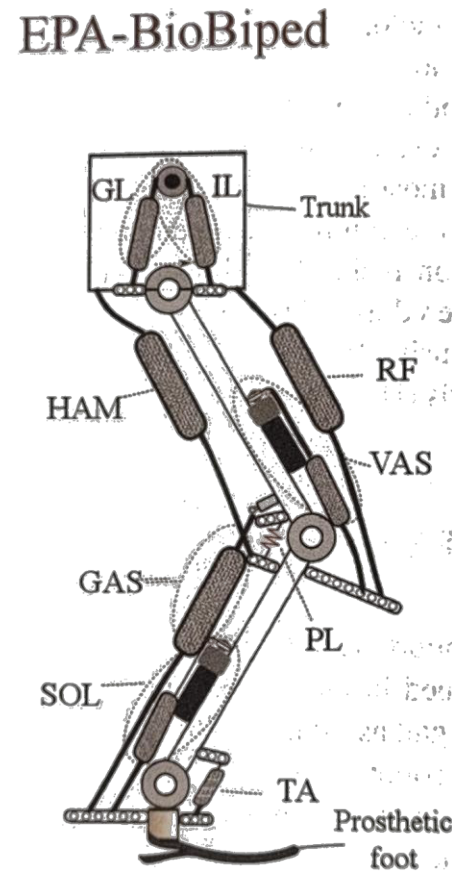
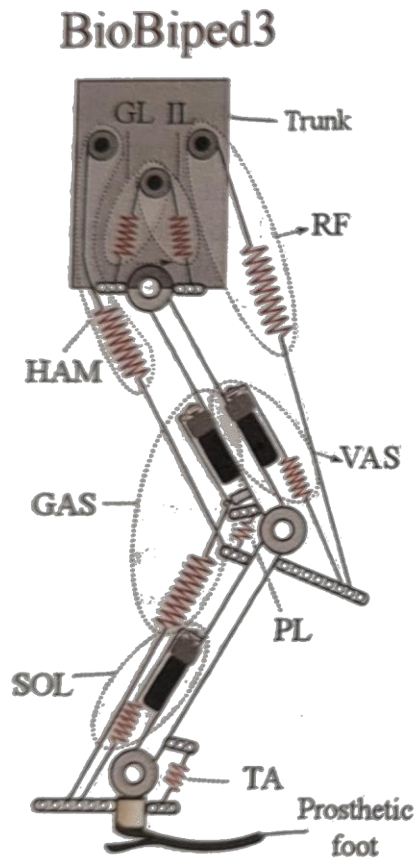
- Parallel 并联式



Series Elastic Actuators – SEA



Hybrid Electric-Pneumatic SEA

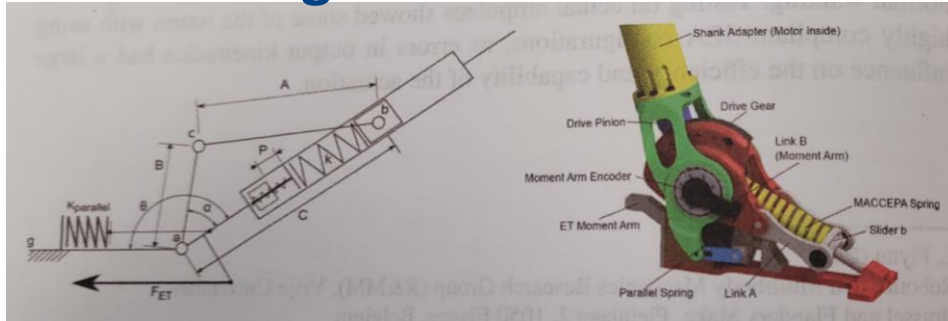


- SEA 被 EPA（电-气动执行器）替代
- SEA are replaced with EPA (Electric-Pneumatic Actuators)
- Passive springs are replaced with PAM (Pneumatic Artificial Muscles)
- 被动弹簧被 PAM（气动人工肌肉）替代

SEA-PEA 示例: CYBERLEG 假肢

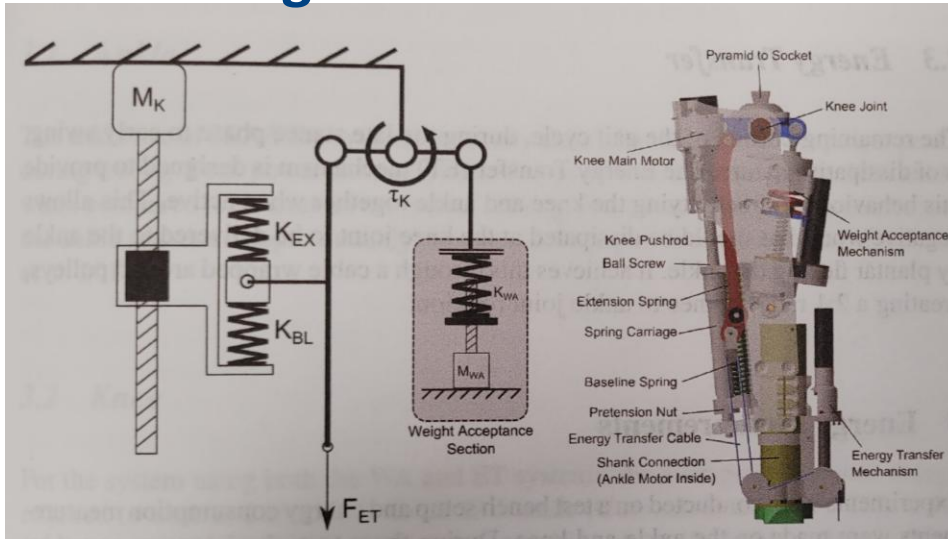
SEA-PEA example: CYBERLEG prosthesis

Ankle design 踝关节设计:



- A main motive actuator
 - Series elastic linkage
 - Fixed parallel spring
- 主驱动执行器
 - 串联弹性连杆
 - 固定并联弹簧

Knee design 膝关节设计:



- 主 SEA 执行器
- Main SEA actuator
- Parallel Weight Acceptance actuator (actuated spring)



• 并联配重

ch?v=a0BEv-wa36Q&ab_channel=CYBERLEGS

• 接受腔执行器（驱动弹簧）

L. Flunn, "Compliant actuation for transfemoral prostheses: the CYBERLEG prosthesis", in: Sharbafi, T. Verstraten, P.P. Pott (Ed.s) *Novel Bioinspired Actuator Designs for Robotics*

Summary

Questions ?

- Introduction to interaction control
- Strategies for controlling interaction in a robot:
 - Impedance/stiffness/admittance/force control
 - Learning-based control
 - Elastic actuators
- 交互控制导论
- 机器人交互控制策略：
 - 阻抗 / 刚度 / 导纳 / 力控制
 - 基于学习的控制
 - 弹性执行器